



Lecture 4:

Multimodal Representation

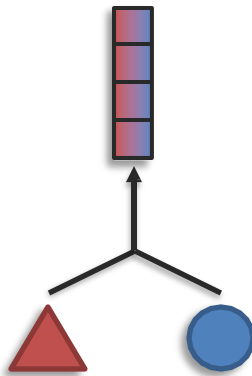
Coordination

Email: idanschwartz@gmail.com or
idan.schwartz@biu.ac.il

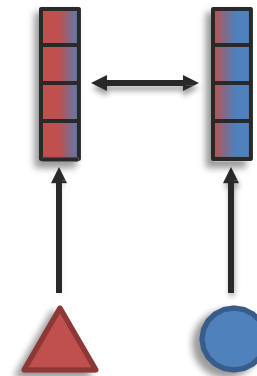
Challenge 1: Representation

Definition: Learning representations that reflect cross-modal interactions between individual elements, across different modalities

Fusion

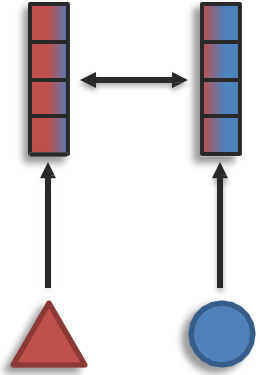


Coordination



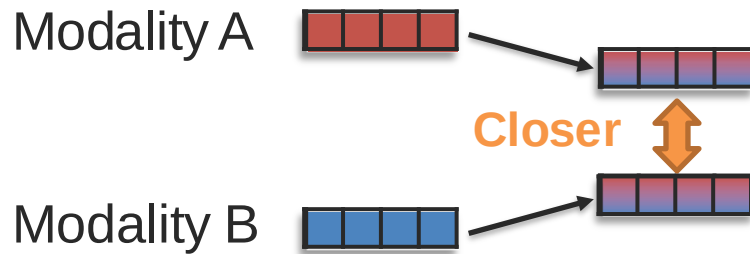
Today

Learning Representation Coordination

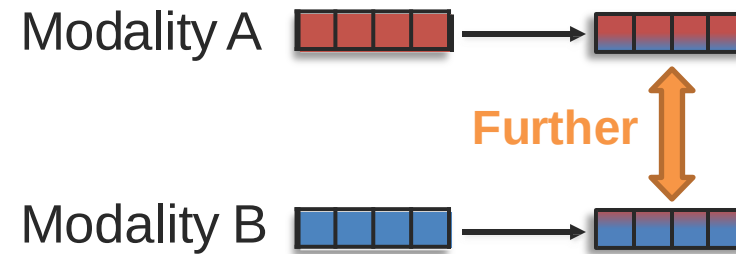


Definition: Learn multimodally-contextualized representations that are coordinated through their cross-modal interactions

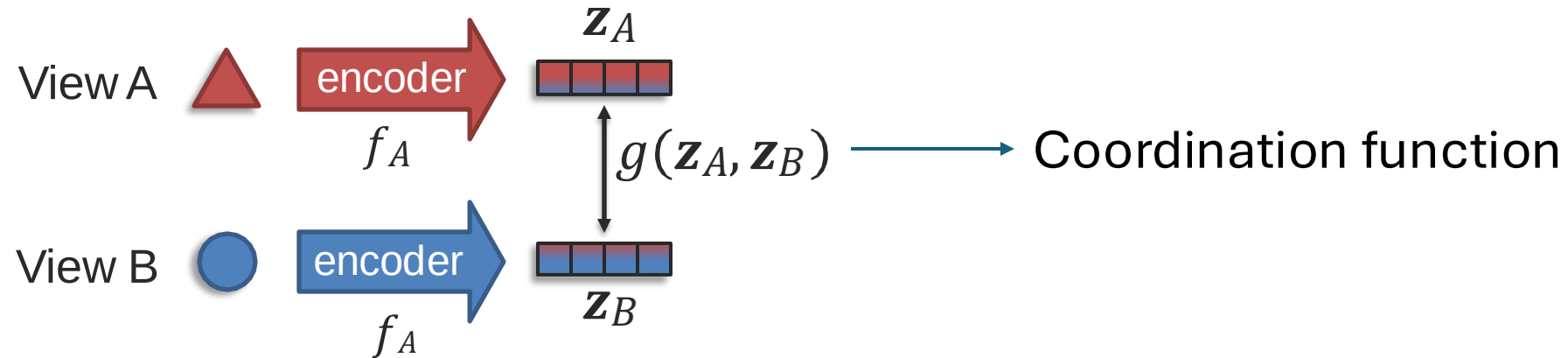
Strong Coordination:



Partial Coordination:



Coordination Function



Cosine similarity:

$$g(z_A, z_B) = \frac{z_A \cdot z_B}{\|z_A\| \|z_B\|}$$

Strong coordination!

Canonical Correlation Analysis (CCA)

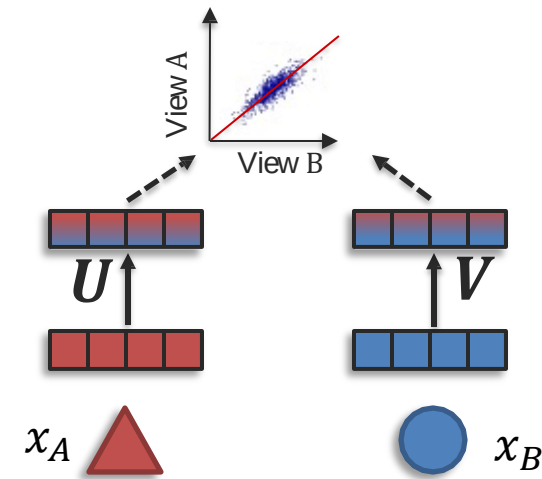
- Learn two linear projections, one for each view, that are maximally correlated:

$$\text{minimize}_{V,U} \quad \frac{1}{N} \sum_{\ell=1}^N \|Ux_A^\ell - Vx_B^\ell\|_2^2$$

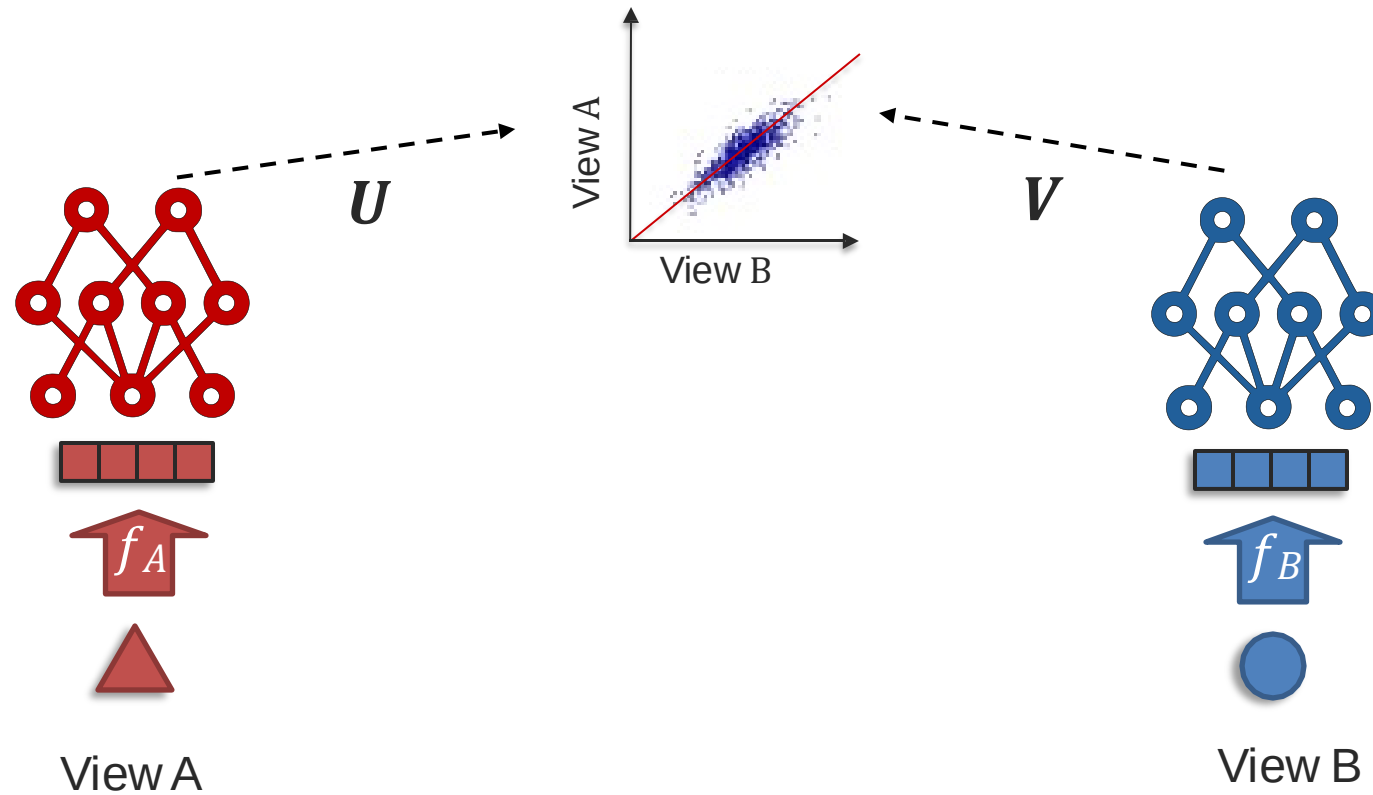
$$\text{s.t.} \quad \frac{1}{N} \sum_{\ell=1}^N Ux_A^\ell (x_A^\ell)^\top (U)^\top = I,$$

$$\frac{1}{N} \sum_{\ell=1}^N Vx_B^\ell (x_B^\ell)^\top (V)^\top = I$$

- N is the dataset size

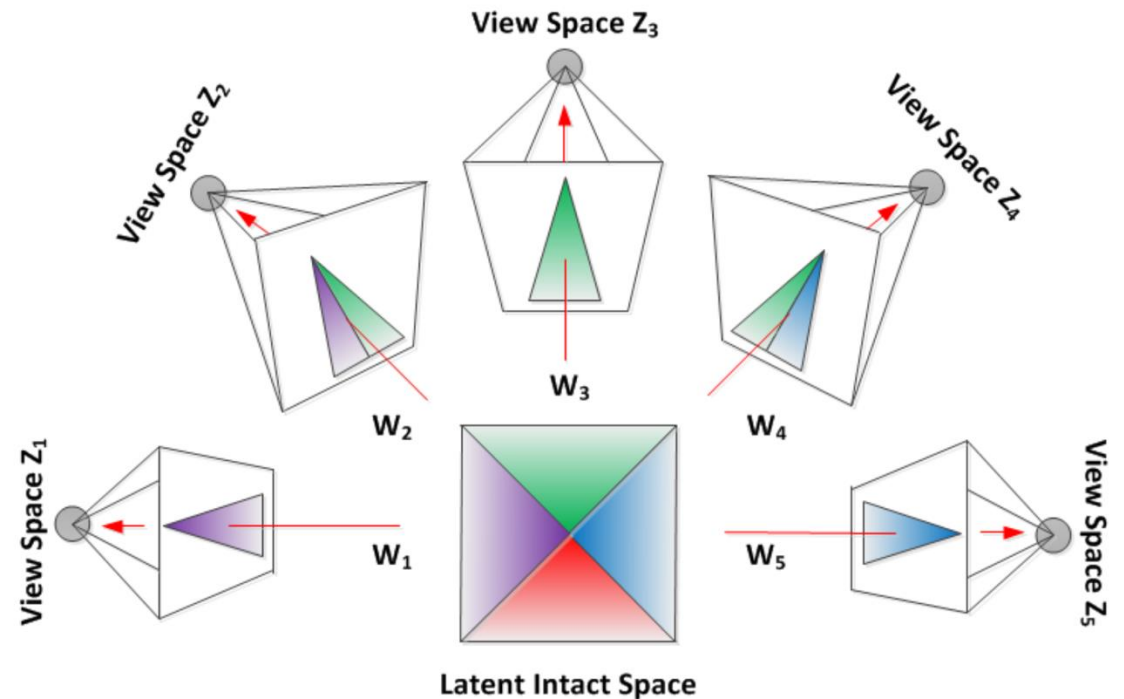


Deep Canonically Correlated Autoencoders (DCCAE)

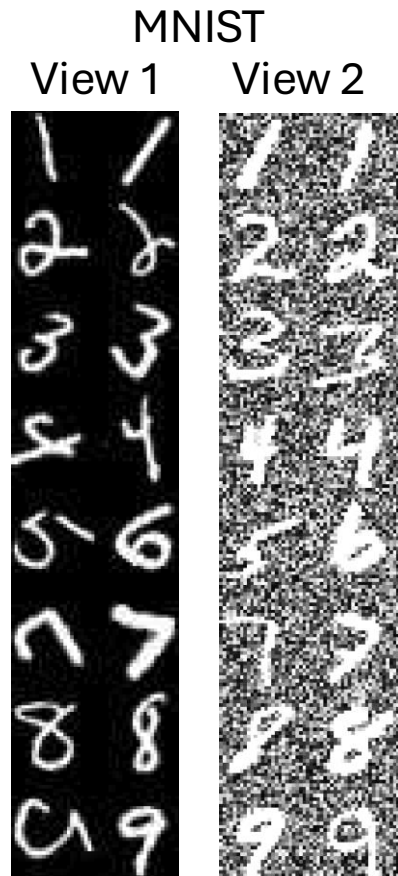


Multi-view latent space

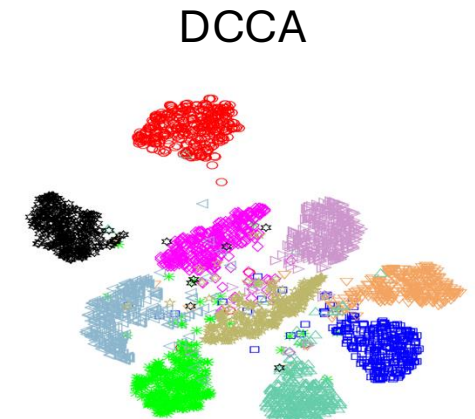
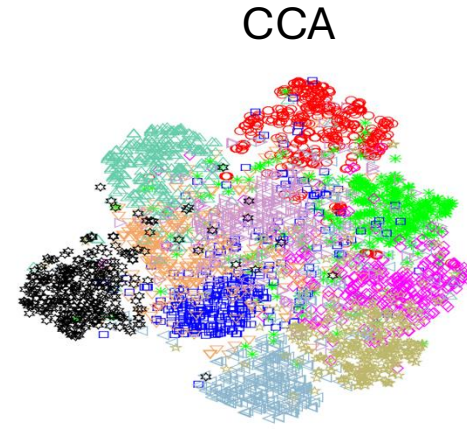
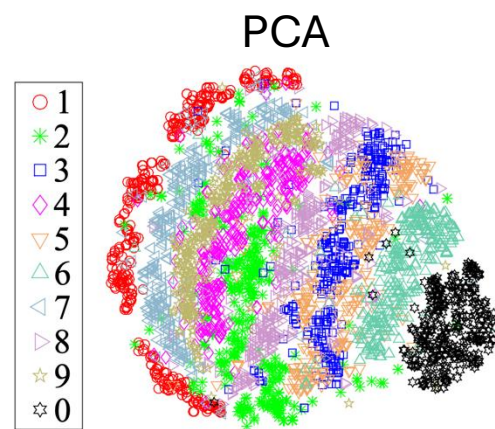
- Given multiple views of the same object
- The views are partial (and possibly degenerated) representations of the intact representation
- There is an “intact” representation which is complete and not damaged



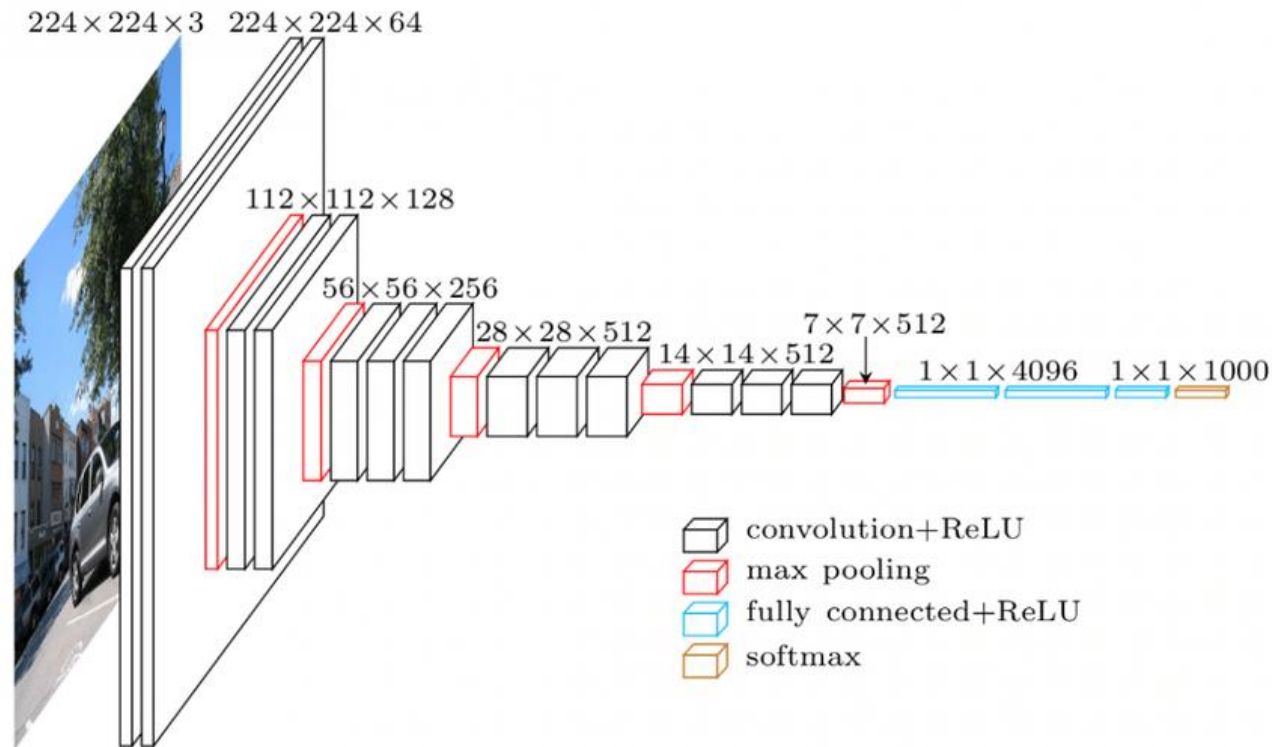
Improve representations with synthetic augmented views



Clustering (no labels):



Recall: Representation learning via image classification



VGG (2014)

Representation learning, challenges



Detected concepts:

Cow: 0.99, Pasture: 0.99, Grass: 0.99,
No Person: 0.98, Mammal: 0.98



Detected concepts:

No Person: 0.99, Water: 0.98, Beach:
0.97, Outdoors: 0.97, Seashore: 0.97

Representation learning, challenges

- ResNet-50 with ImageNet supervision

Labels Fraction	1%	10%	100%
Top1 Acc	25.4%	56.4%	78.9%
Top-5 Acc	48.4%	80.4%	94.2%

- 10% training data = 100,000 samples
- High workload for annotations

Multiview representation

- Multiview representation aims to extract mutual information across multiple views



SimCLR: A framework for data augmentations



(a) Original



(b) Crop and resize



(c) Crop, resize (and flip)



(d) Color distort. (drop)



(e) Color distort. (jitter)



(f) Rotate $\{90^\circ, 180^\circ, 270^\circ\}$



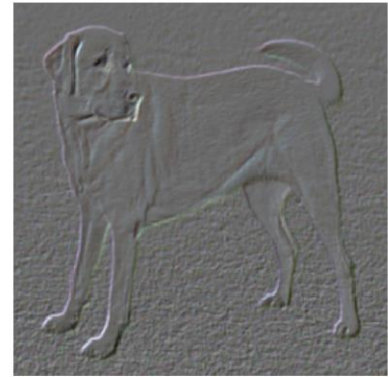
(g) Cutout



(h) Gaussian noise



(i) Gaussian blur



(j) Sobel filtering

Contrastive loss

- Contrastive loss:

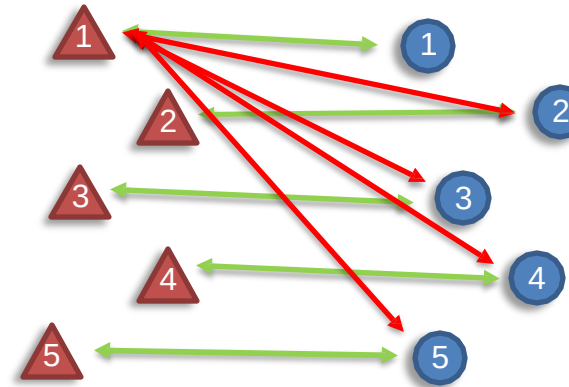
- brings **positive pairs** closer
- pushes **negative pairs** apart

- Simple contrastive loss:

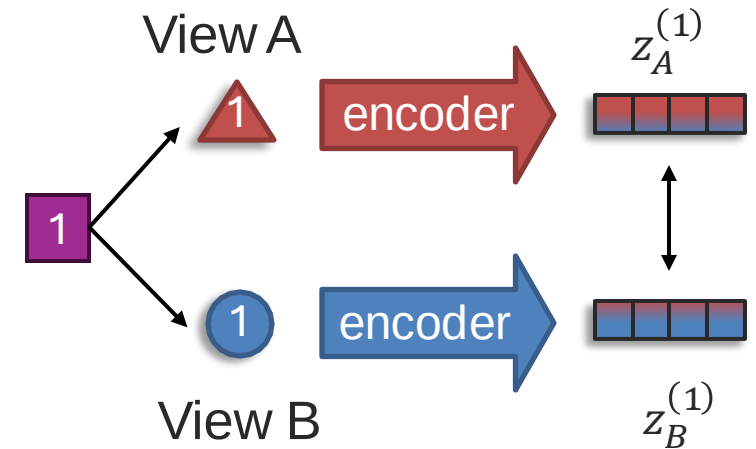
$$\mathcal{L} = \max \left\{ \alpha + \underline{\text{sim} \left(z_A^{(i)}, z_B^{(i)} \right)} - \underline{\text{sim} \left(z_A^{(i)}, z_B^{(j)} \right)}, 0 \right\}$$

- Similarity is often measured with cosine operator

Multiview data: { ,  }



 Positive pairs
 Negative pairs



SimCLR - animation

SimCLR results

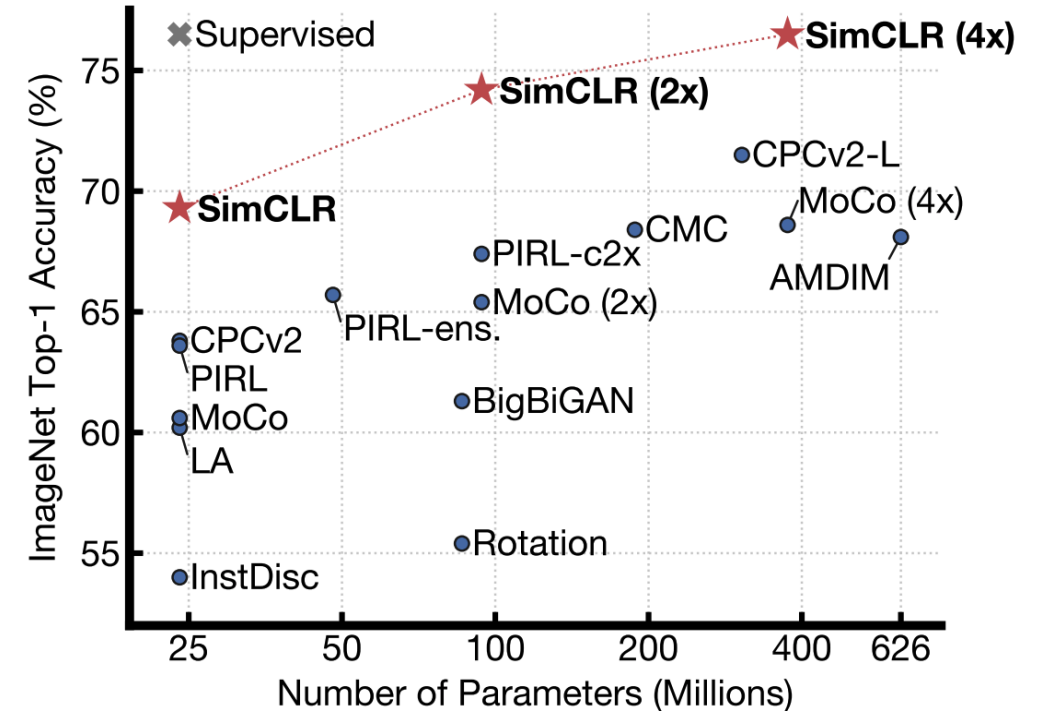
- Supervised:

Labels Fraction	1%	10%	100%
Top1 Acc	25.4	56.4	78.4
Top-5 Acc	48.4	80.4	94.2

- Self-Supervised + Linear Classifier w\ ImageNet:

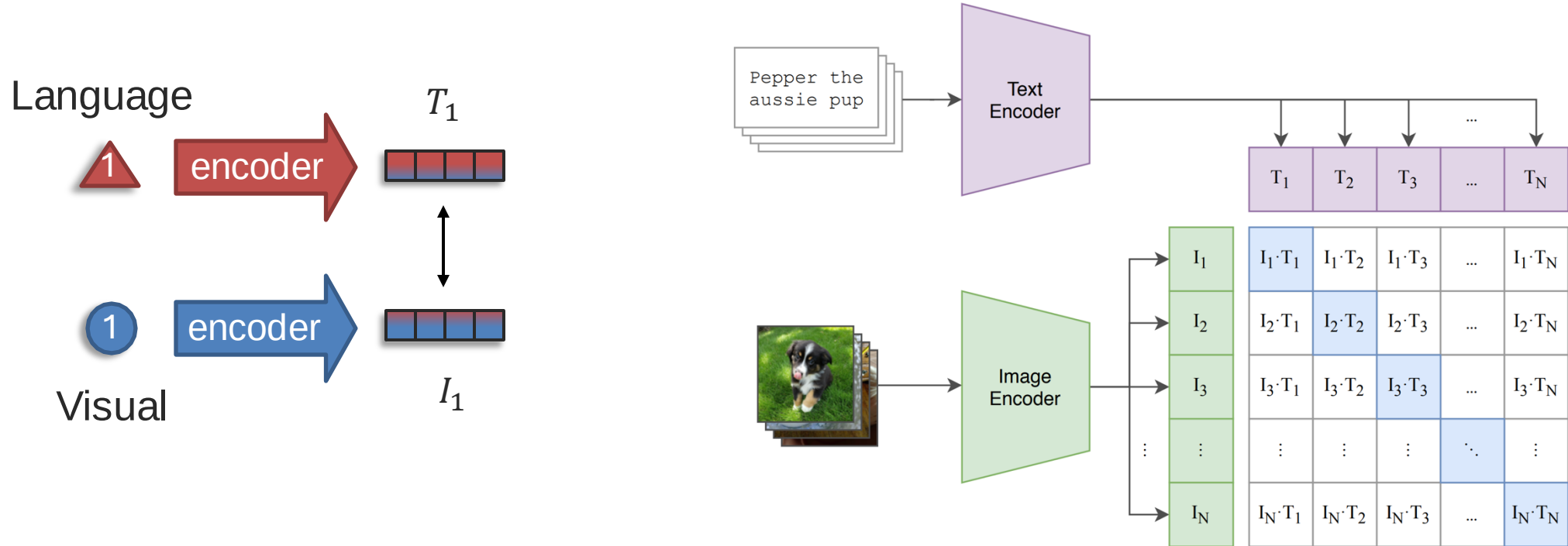
Labels Fraction	1%	10%	100%
Top1 Acc	48.3	65.6	69.3
Top-5 Acc	75.5	87.8	89.0

Self-Supervised features are as useful as supervised features:



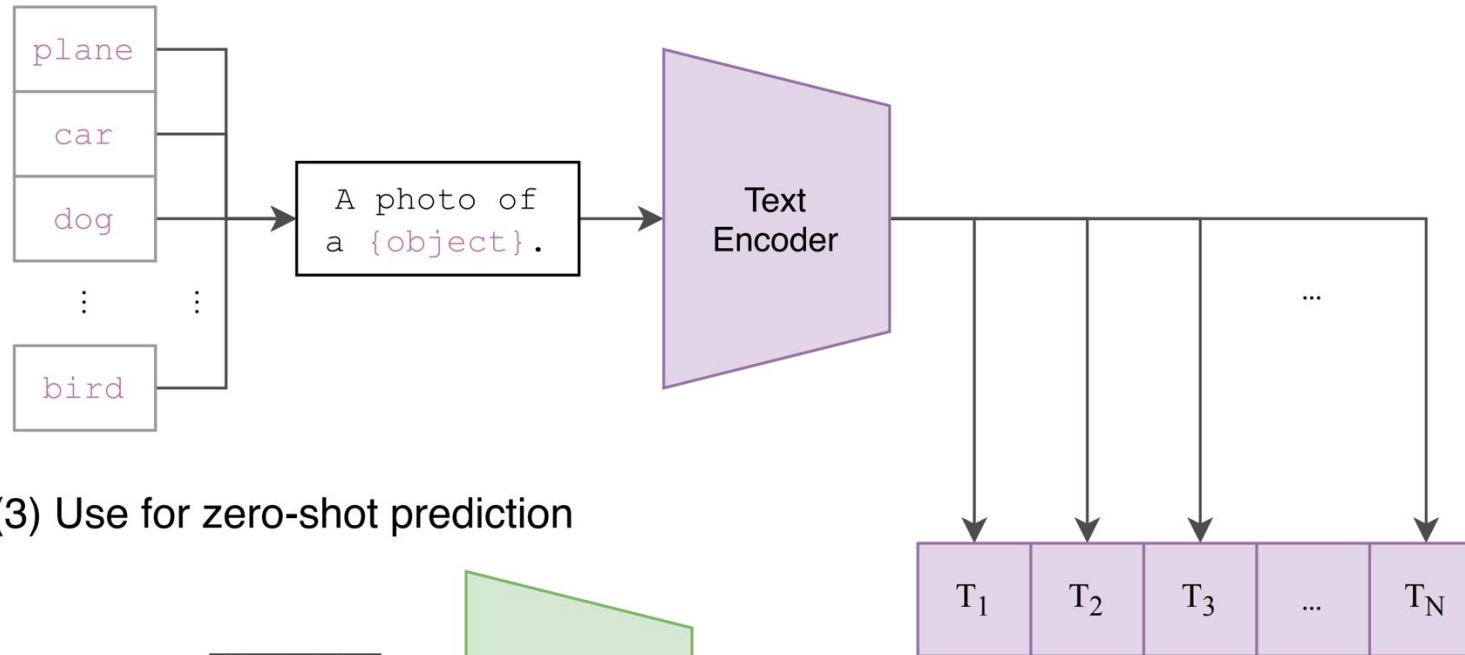
Contrastive Language–Image Pre-training

- Dataset of 400 million (image, text) pairs collected from the internet

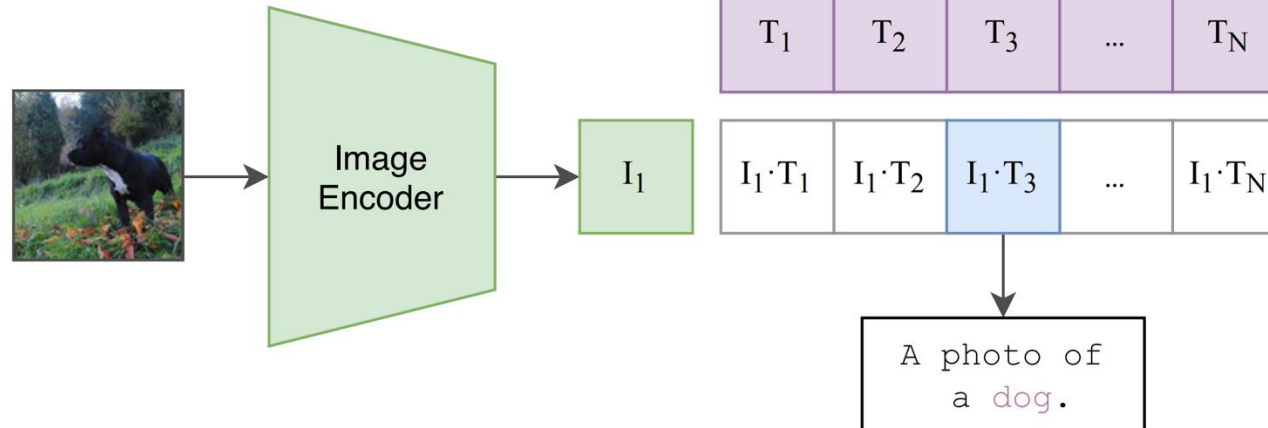


Zero-shot classification

(2) Create dataset classifier from label text

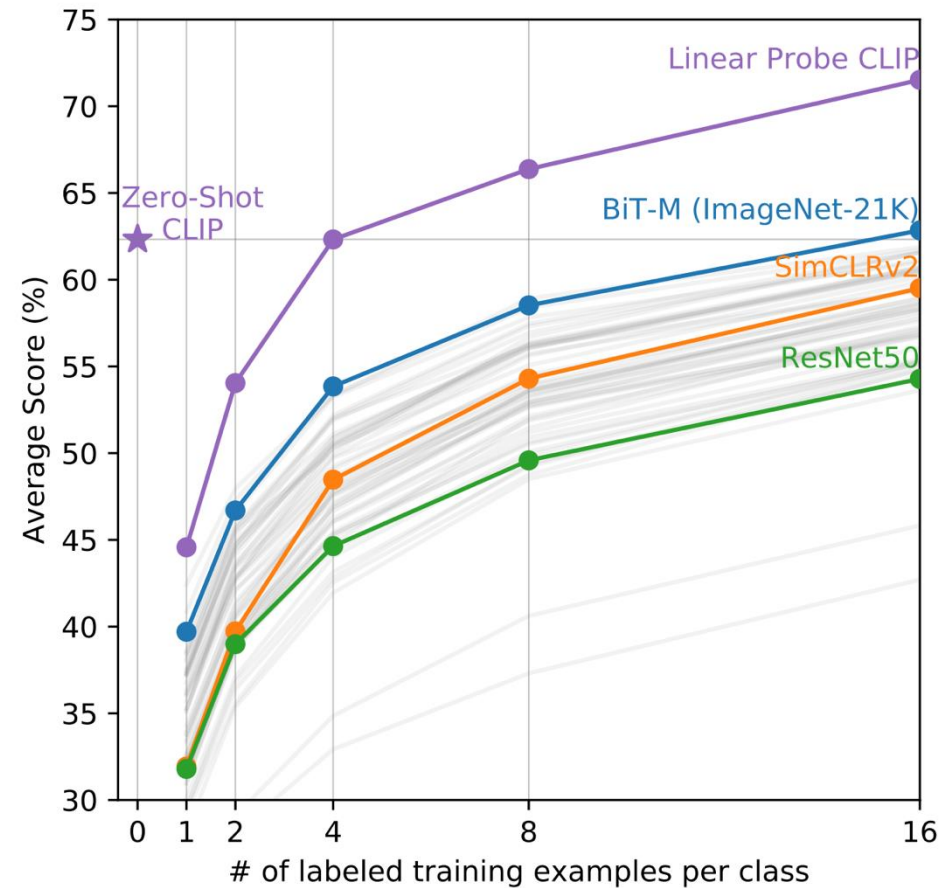


(3) Use for zero-shot prediction

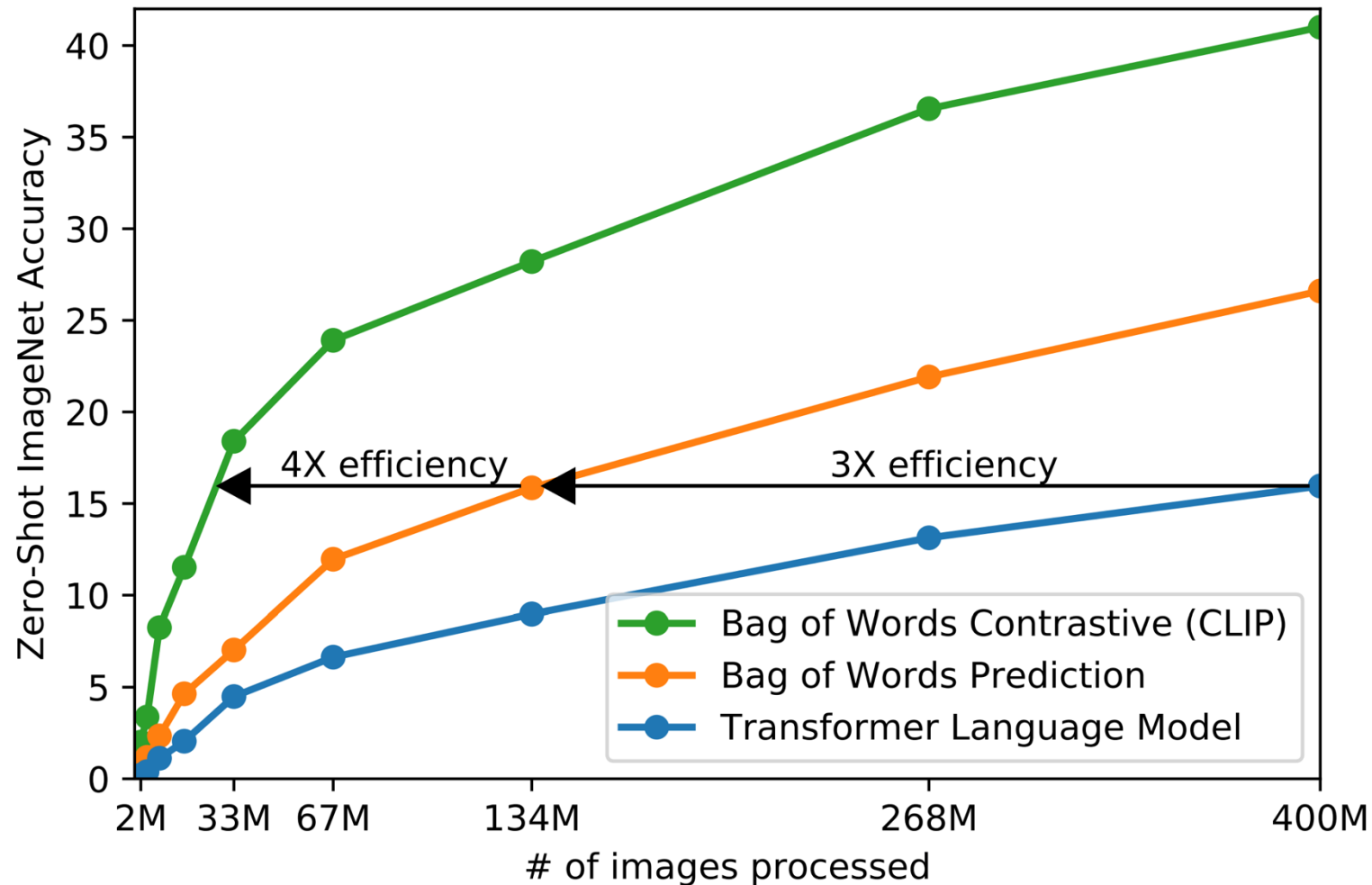


Much better representations for transfer learning

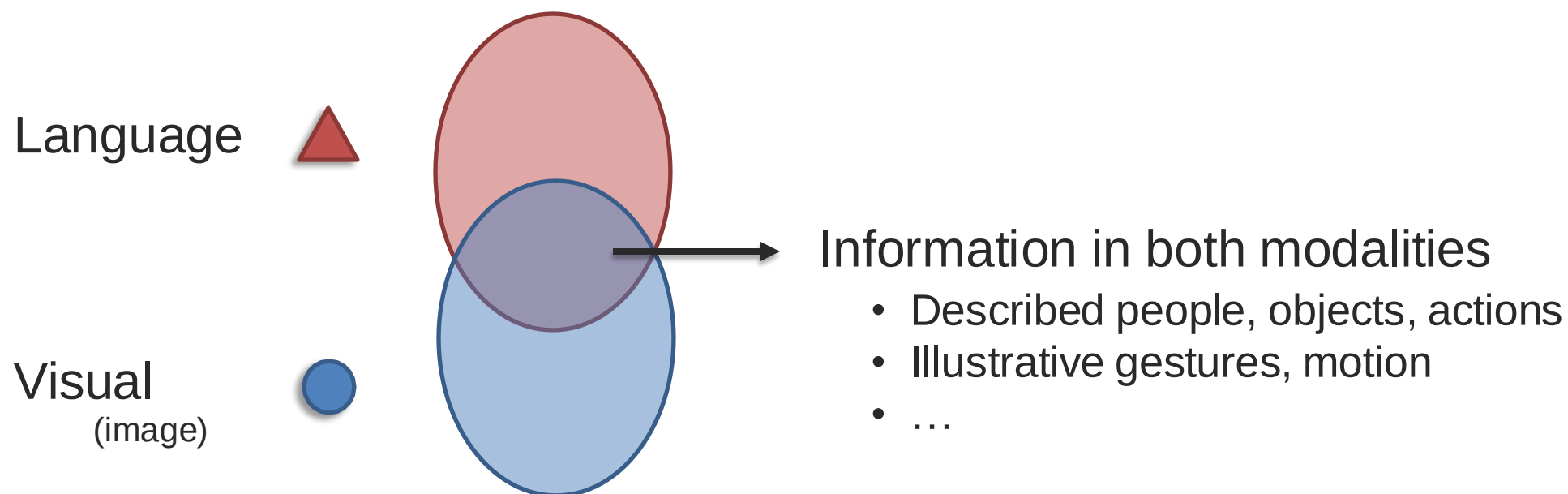
- CLIP representations surpass previous on few-shot transfer learning
- Zero-shot CLIP outperforms 2-shot linear probes



Contrastive learning efficiently learns representations



What contrastive loss learns?

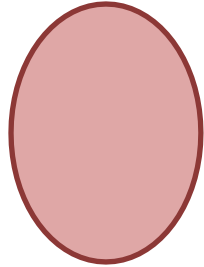


Information and Entropy – Information Theory

Language



x



How much information in the modality?

Information Theory (Shannon, 1948)

Main intuition: “Information value” of a communicated message x depends on how random its content is

x : “1,1,1,1,1,1,1,1,1,1,1,1,1”

➡ Not very random... So, low information

x : “0,1,0,1,0,0,1,1,1,0,0,1”

➡ More random... So, higher information

Information content $I(x)$

$$I(x) \sim \frac{1}{p(x)}$$

$$I(x) = \log\left(\frac{1}{p(x)}\right) = -\log(p(x))$$

Information and Entropy – Information Theory

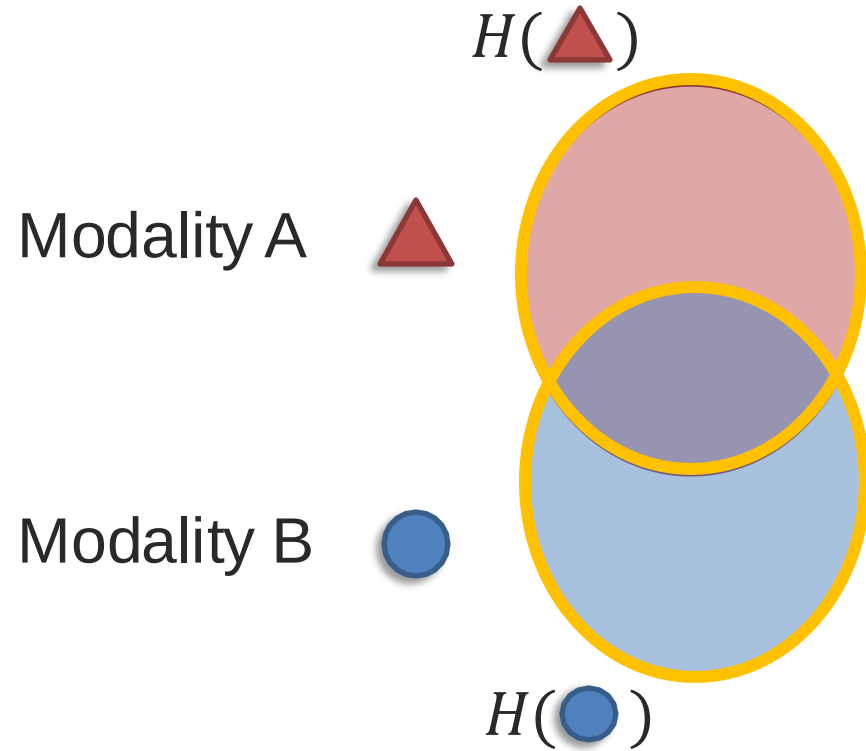
Information content $I(X) = -\log(p(X))$

➡ For discrete alphabet $\mathcal{X}$, i.e., X is a discrete random variable

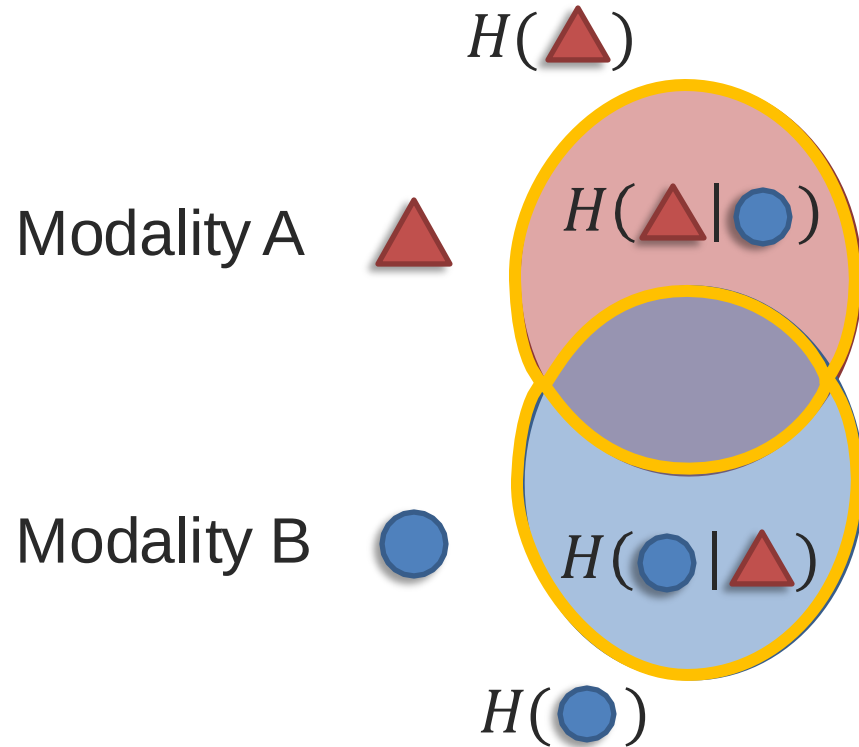
Entropy: weighted average of all possible outcomes from $\mathcal{X}$

$$H(X) = \mathbb{E}[I(X)] = \mathbb{E}[-\log(p(X))] = - \sum_{x \in \mathcal{X}} p(x) \log(p(x))$$

Entropy with Two Modalities



Entropy with Two Modalities



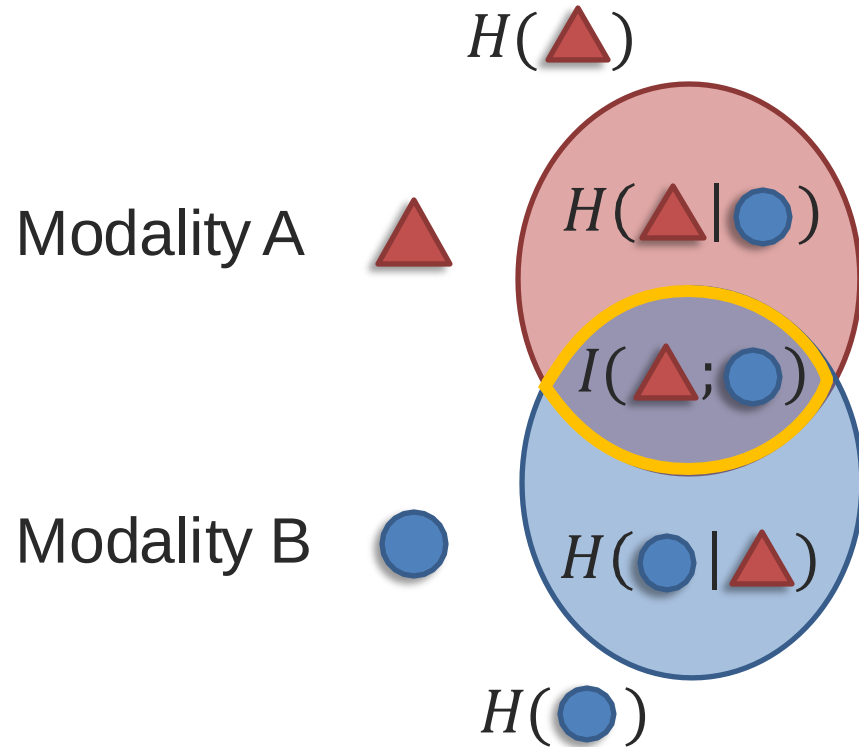
Conditional entropy $H(Y|X)$

$$\begin{aligned} H(Y|X) &= -\mathbb{E}[\log(p(y|x))] \\ &= -\mathbb{E}\left[\log\left(\frac{p(x,y)}{p(x)}\right)\right] \end{aligned}$$

If X and Y independent, $H(Y|X) = H(Y)$

If X fully determines Y , then $H(Y|X) = 0$

Entropy with Two Modalities



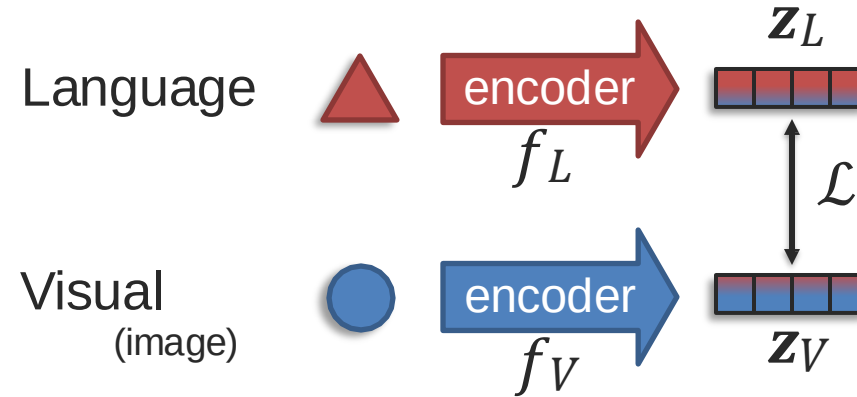
Mutual information $I(X; Y)$

$$I(X; Y) = H(X) - H(X|Y)$$

$$= \mathbb{E} \left[\log \left(\frac{1}{p(x)} \right) + \log \left(\frac{p(x, y)}{p(y)} \right) \right]$$

$$= \mathbb{E} \left[\log \left(\frac{p(x, y)}{p(x)p(y)} \right) \right]$$

CLIP's contrastive loss



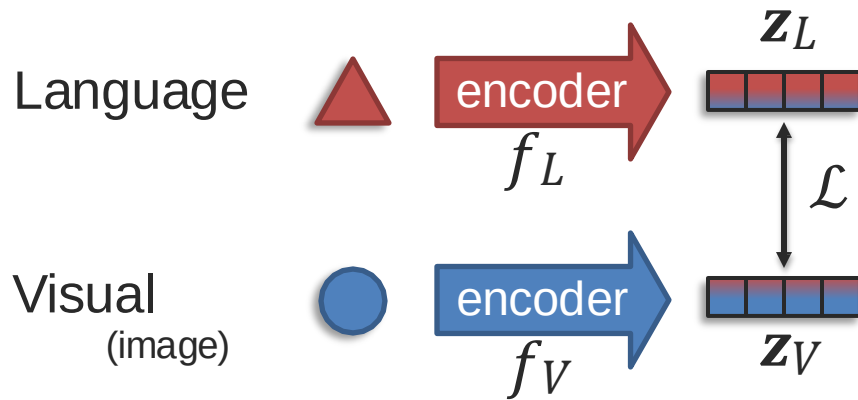
Popular contrastive loss: InfoNCE

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^i)}{\sum_{j=1}^N \text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^j)}$$

Annotations for the equation:

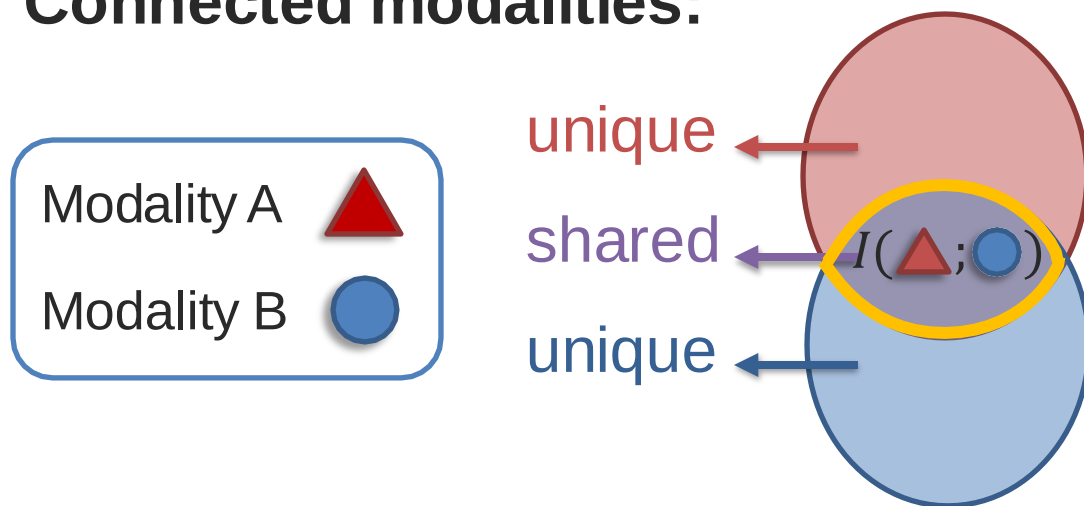
- A green arrow points from the text "positive pairs" to the numerator $\text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^i)$.
- An orange arrow points from the text "Similarity function can be cosine similarity" to the sim function in the denominator.
- A red arrow points from the text "negative pairs and positive pairs" to the denominator $\sum_{j=1}^N \text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^j)$.

Contrastive Learning and Connected Modalities



$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^i)}{\sum_{j=1}^N \text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^j)}$$

Connected modalities:



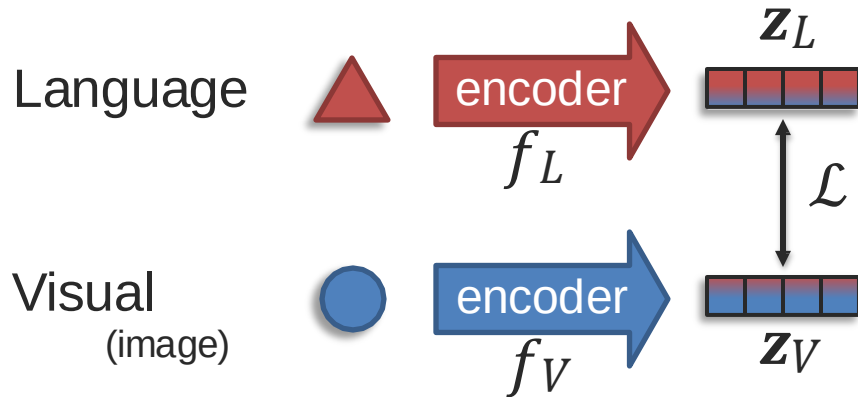
Mutual information $I(X; Y)$

$$\mathbb{E} \left[\log \left(\frac{p(x, y)}{p(x)p(y)} \right) \right]$$



CLIP focuses on shared connections

Contrastive Learning and Mutual Information



InfoNCE:

$$\mathcal{L} = -\mathbb{E} \left[\log \frac{f(x_A^i, x_B^i)}{\sum_{j=1}^N f(x_A^i, x_B^j)} \right]$$

Critic function f is trained to be a binary classifier distinguishing $x_A, x_B \sim p(x_A, x_B)$ vs $x_A, x_B \sim p(x_A) p(x_B)$

InfoNCE/CL:

- 'Captures' mutual information
- Optimizes a lower bound on mutual information

At optimal loss, $f^*(x_A, x_B) = \frac{p(x_A, x_B)}{p(x_A)p(x_B)}$

Plugging f^* back into $\mathcal{L}$ gives:

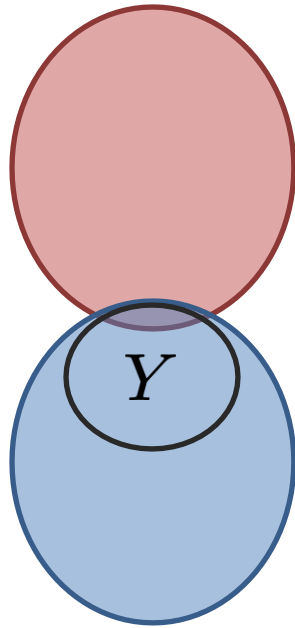
$$\mathcal{L}^* \geq \mathbb{E} \left[\log \frac{p(x_A, x_B)}{p(x_A)p(x_B)} N \right] = -I(X_A, X_B) + \log N$$

In other words:

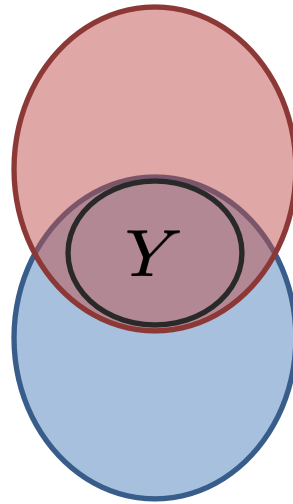
$$I(X_A, X_B) \geq \log N - \mathcal{L}^*$$

How much information should be shared?

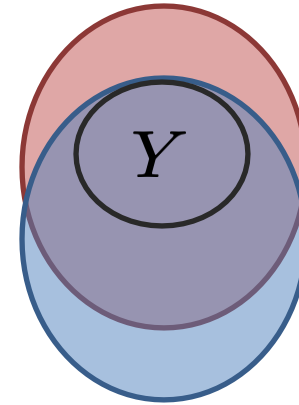
Not enough signal



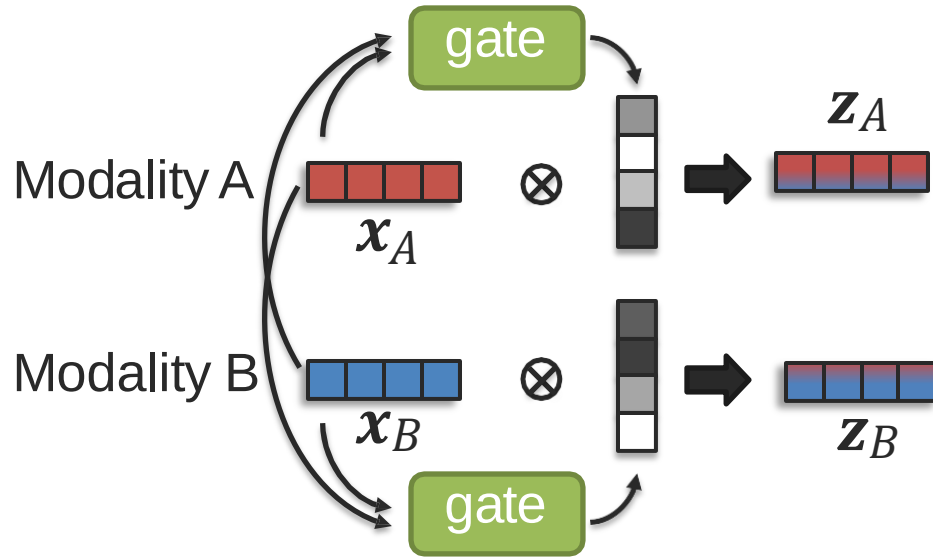
Just right



Too much noise



Gated Coordination



Gated coordination:

$$z_A = g_A(x_A, x_B)x_a$$

$$z_B = g_B(x_A, x_B)x_b$$

Also known as Co-Attention*

*-Hierarchical Question-Image Co-Attention for Visual Question Answering; Lu et al; NIPS'16

Visual question answering



What animal is in the cage?

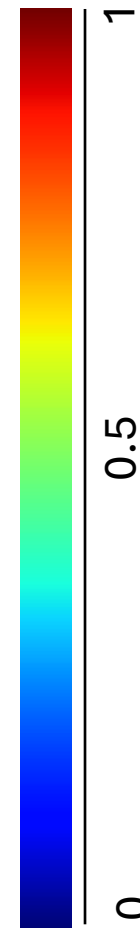
VQA Co-Attention



What **color** is the plastic **container**
inside the cage?

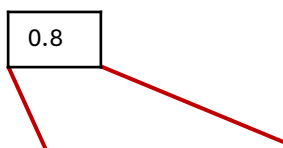


What **animal** is **in the** cage?



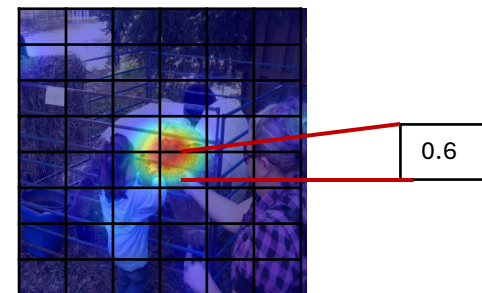
VQA elements

What animal is in the cage?



Elements : words

$$(Q_1, \dots, Q_{15}) \in \mathbb{R}^{15 \times d_Q}$$



Elements: image regions

$$(I_1, \dots, I_{49}) \in \mathbb{R}^{49 \times d_I}$$

VQA Co-Attention

- Given an image I and a question Q
- We infer the amount of attention that should be given to each image-region and a word by computing attention belief b :

$$g_I(i) \propto \exp(b(I, Q, i)), \quad g_Q(i) \propto \exp(b(I, Q, i))$$

- We calculate the attend representation:

$$a_Q = \sum_{i=1..15} g_Q(i) \cdot Q_j \quad ; \quad a_I = \sum_{j=1..49} g_I(i) \cdot I_i$$

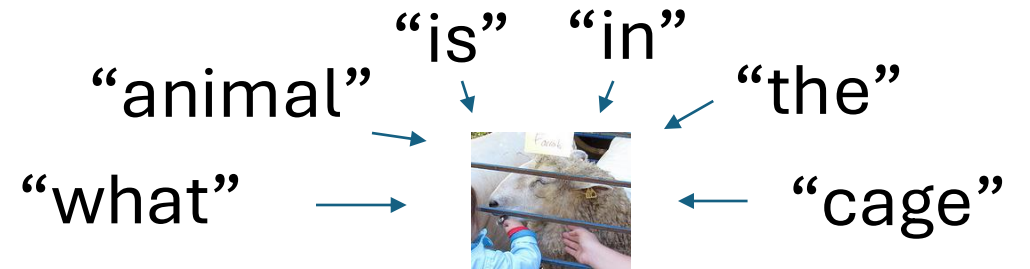
Factorize the attention belief

$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Unary ($\psi_I(i)$): region information

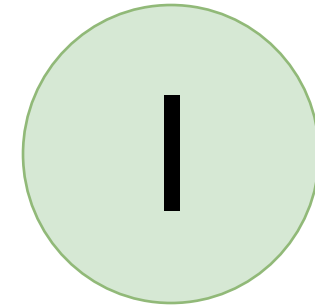
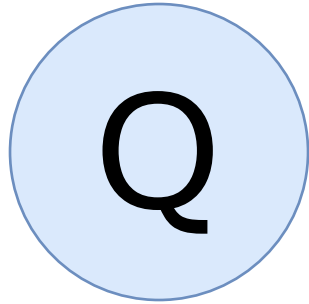


question-to-Image ($\mu_{Q \rightarrow I}(i)$):
interactions of regions with words



Factor Graph Attention

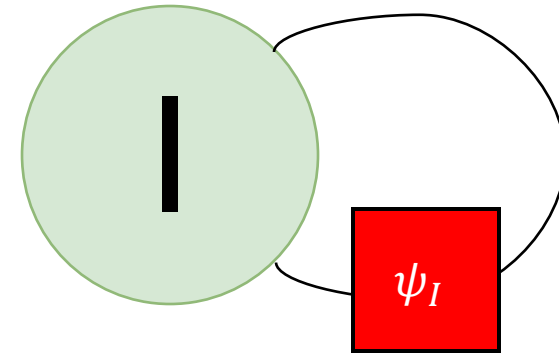
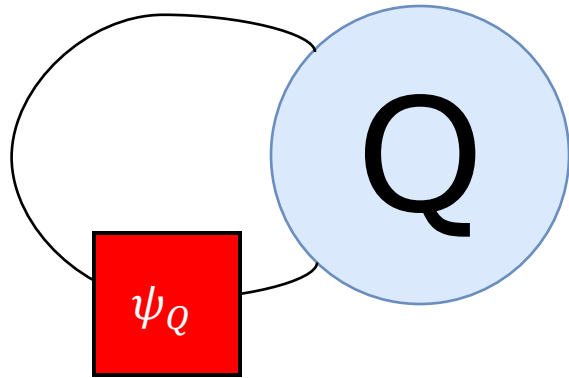
- Each modality is treated as a node



What animal is in the cage?



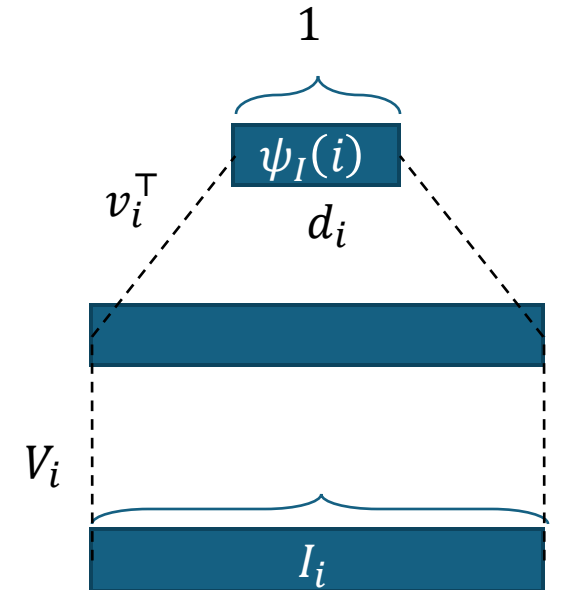
Unary factor



For each image region:

$$\psi_I(i) = v_i^\top \text{relu}(V_i I_i)$$

Where weights $v_i^\top \in \mathbb{R}^{d_I \times 1}$ and $V_i \in \mathbb{R}^{d_I \times d_I}$



$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Unary attention visualize

- Entity information reacts to strong features across the dataset
- Can reveal dataset **bias**



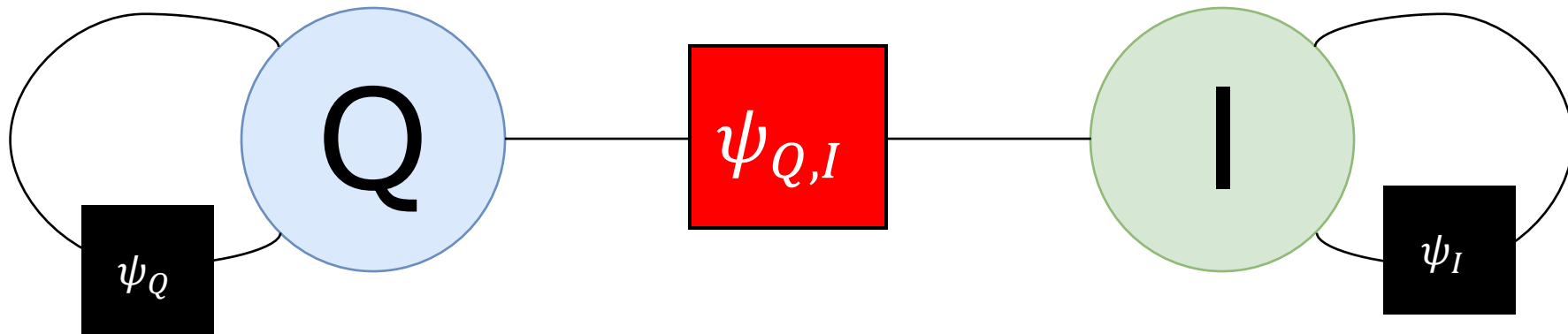
What animal is in the cage?



What color is the plastic container inside the cage?

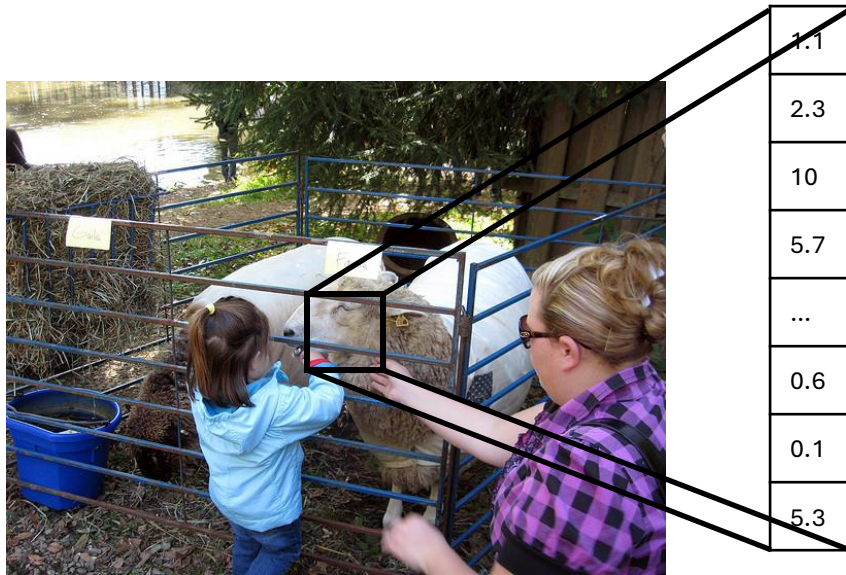
$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Pairwise Interaction factor



$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Interactions via dot-product



•

3.2	5.1	9	6	...	0.1	0.4	7.8
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 = 6.5

What animal is in the cage?

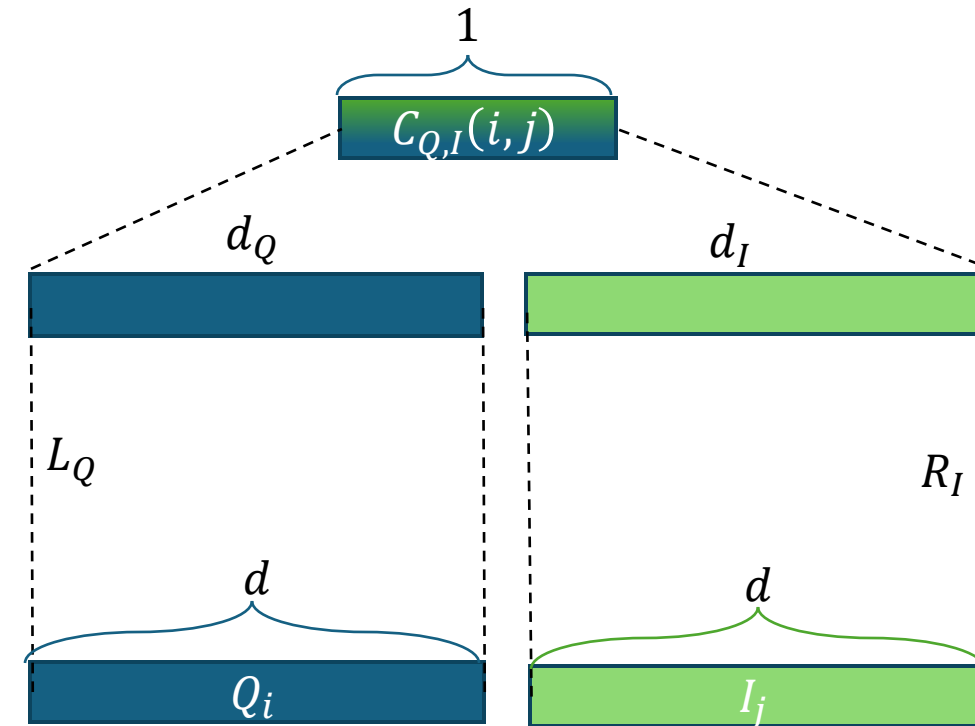
$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

The interaction is learned

- For each two modalities

$$C_{Q,I}(i,j) = \left(\frac{L_Q Q_i}{\|L_Q Q_i\|} \cdot \frac{R_I I_j}{\|R_I I_j\|} \right)$$

- Where weights $L_q \in \mathbb{R}^{d_Q \times d}$ and $R_i \in \mathbb{R}^{d_I \times d}$



$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Pairwise interaction matrix



What animal
is in the cage?

$$C_{Q,I}(i,j) = \left(\frac{L_Q Q_i}{\|L_Q Q_i\|} \cdot \frac{R_I I_j}{\|R_I I_j\|} \right)$$

			
What	1.2	2.3	1.7
animal	-0.4	6.5	4.2
is	1.3	1.2	0.6

$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Message from question to image

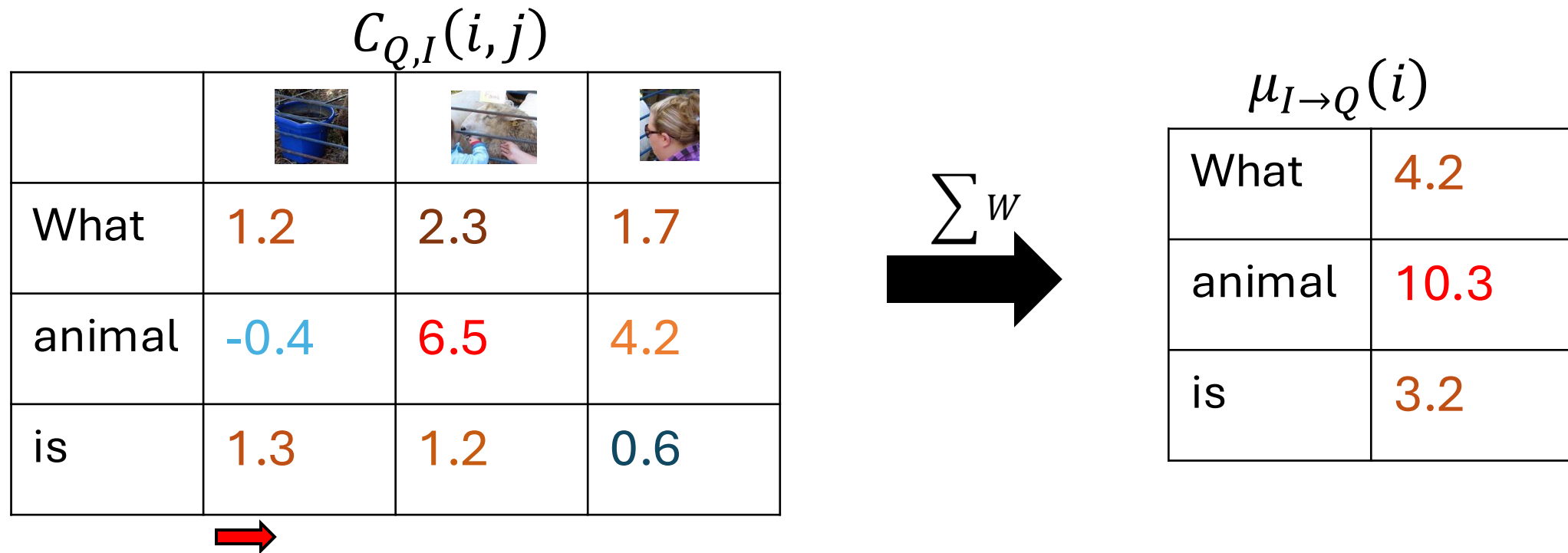
- Messages via weighted marginalization: $\mu_{Q \rightarrow I}(i) = \sum_{Q_j} W_{I,i,j} C_{Q,I}(i,j)$

	$C_{Q,I}(i,j)$				$\mu_{Q \rightarrow I}(i)$	
						2.1
What	1.2	2.3	1.7	$\sum^w$ 		9
animal	-0.4	6.5	4.2			6.5
is	1.3	1.2	0.6			

$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Message from image to question

- Messages via weighted marginalization: $\mu_{Q \rightarrow I}(i) = \sum_{I_j} W_{I,i,j} C_{Q,I}(j, i)$



$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Message's visualization

- Messages reacts to highly word-correlated regions



What **animal** is in the cage?

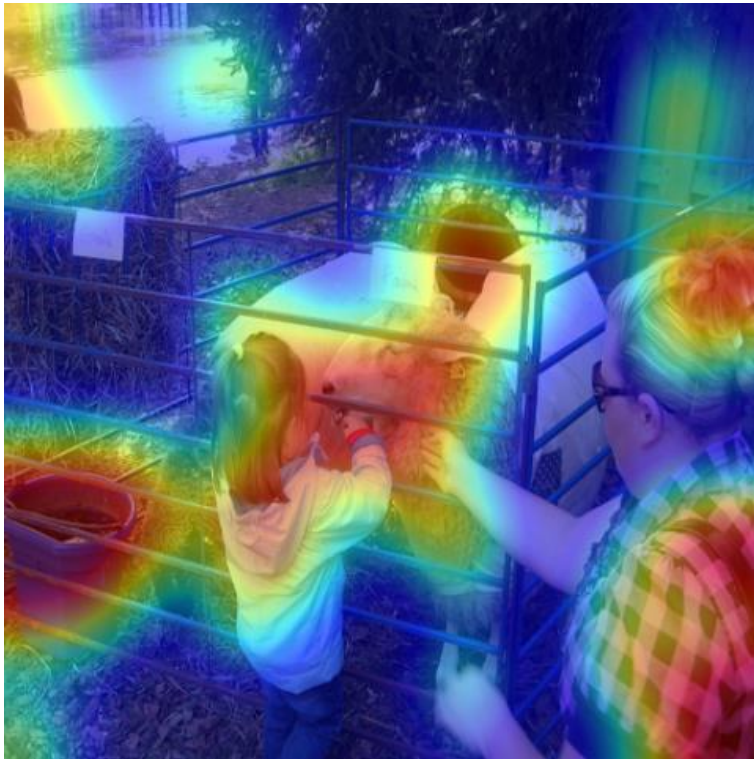


What **color** is the plastic container inside the cage?

$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Unary potential, pairwise messages and belief

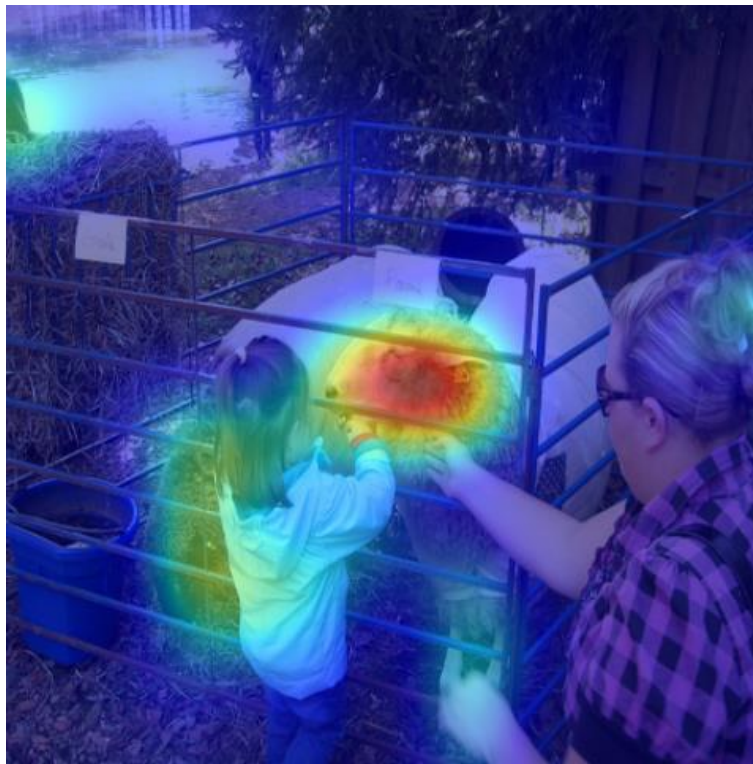
$$\psi_I(i)$$



What **animal** is in the cage?

$$\psi_Q(i)$$

$$\mu_{Q \rightarrow I}(i, j)$$



What **animal** is in the cage?

$$\mu_{I \rightarrow Q}(i, j)$$

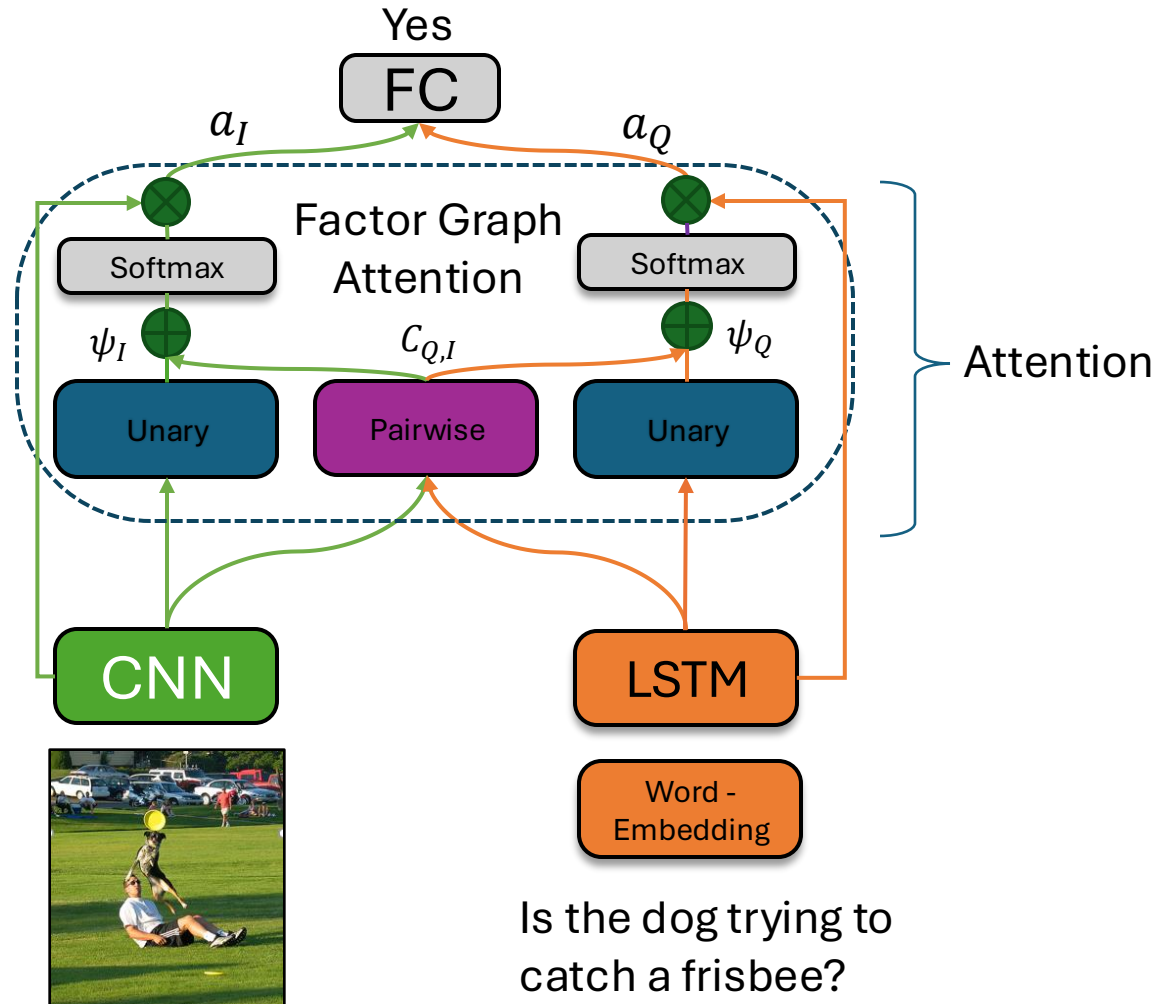
$$b_I(i)$$



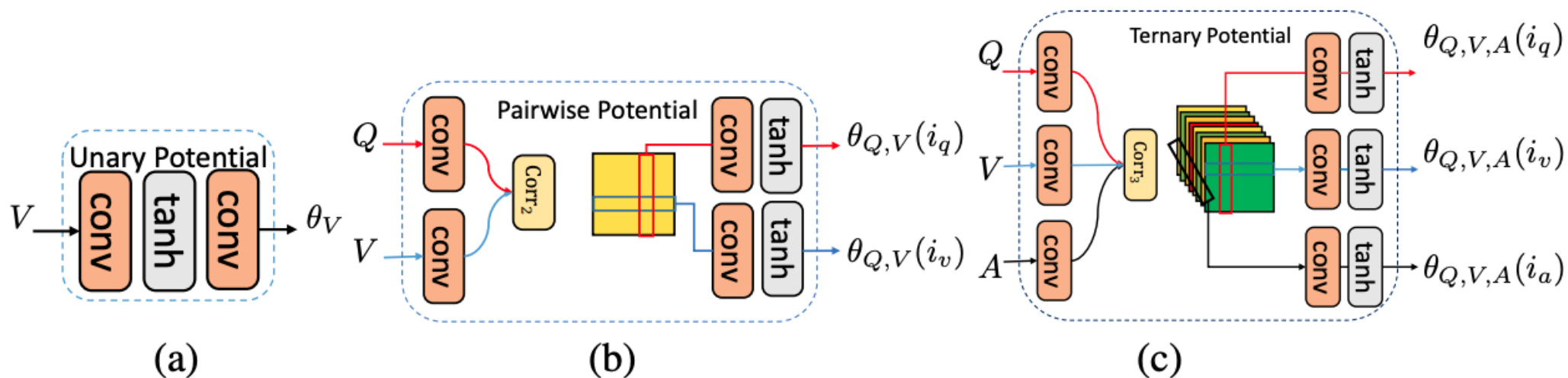
What **animal** is in the cage?

$$b_Q(j)$$

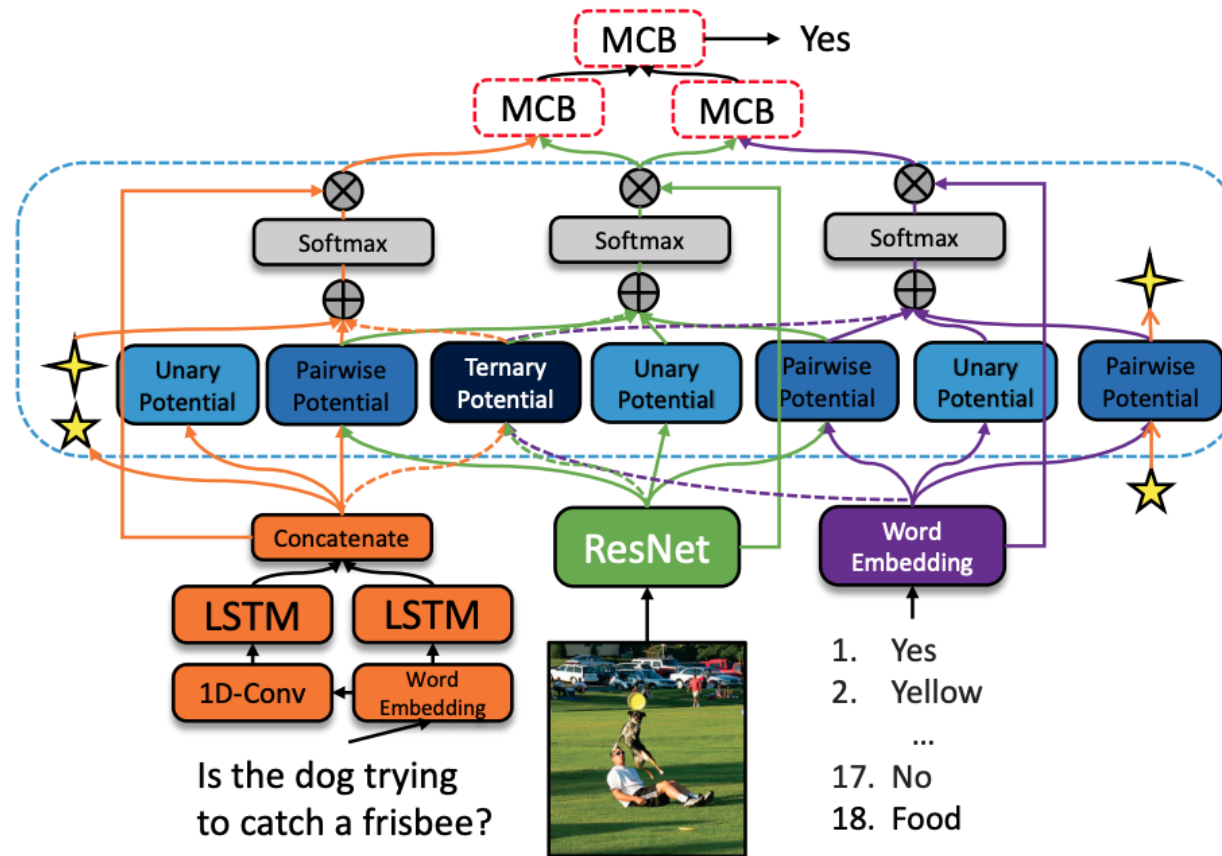
Strong VQA baseline



High-order attention belief



State-of-the-art MC-VQA



Many utilities: Visual Dialog

<i>Image</i>	<i>Caption</i>	<i>Answers</i>
	a group of kids around a table with a cake .	1. Green 2. Yes 3. Not Sure 4. No 5. Yellow 6. 5 Candles ... 99. Five 100. Chocklate
<i>Question</i> Is this outdoors?	<i>History</i> Q: How many kids? A: 5 Q: Is this birthday party? A: Yes Q: Any adults? A: lady lighting candle	

Visual Dialog – Many gate attentions

Question

are kids wearing hats ?

History

Q: how many kids ?

A: 5

Q: is this birthday party?

A: yes

Q: any adults ?

A: lady lighting candle

Q: is this outdoors ?

A: not very sure

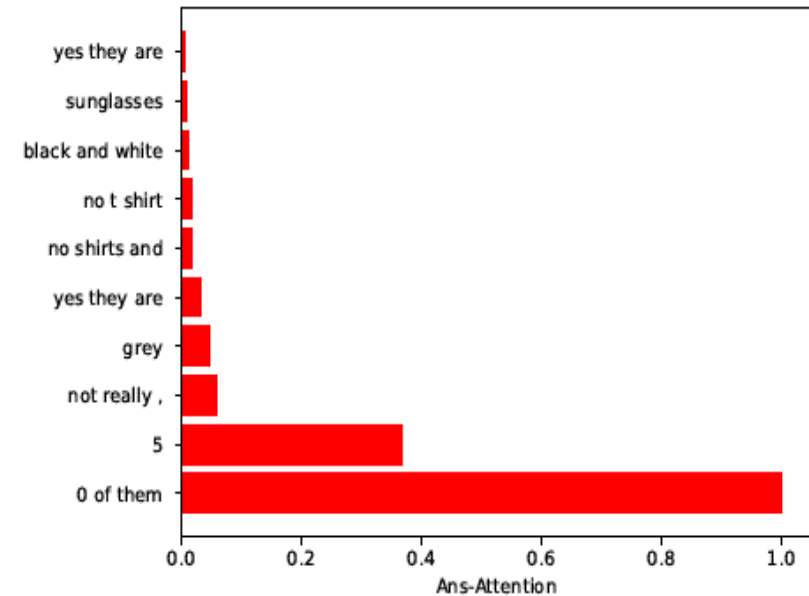
Q: is table round ?

A: no



Caption

a group of kids around a table with a cake .



All gates depends on each other

Question

does cake have candles ?

History

Q: how many kids ?

A: 5

Q: is this birthday party ?

A: yes

Q: any adults ?

A: lady lighting candle

Q: is this outdoors ?

A: not very sure

Q: is table round ?

A: no

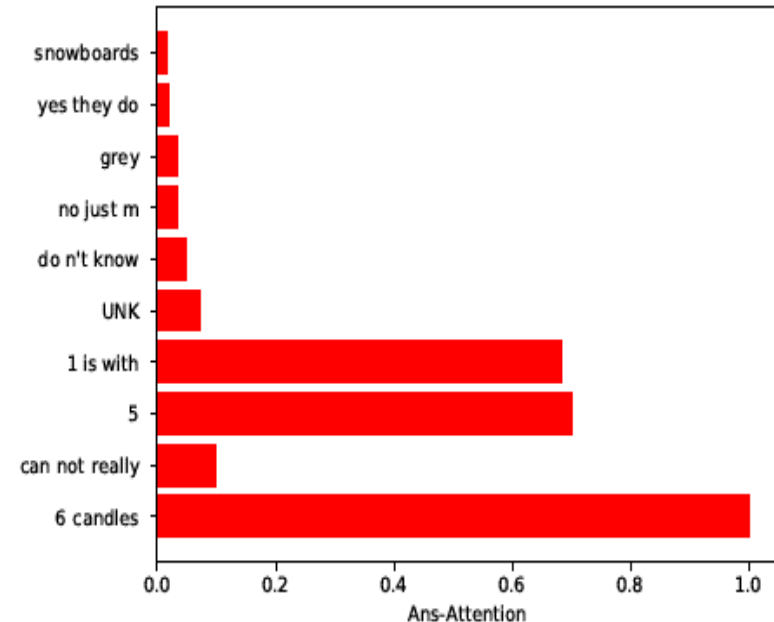
Q: are kids wearing hats ?

A: 0 of them wearing hats



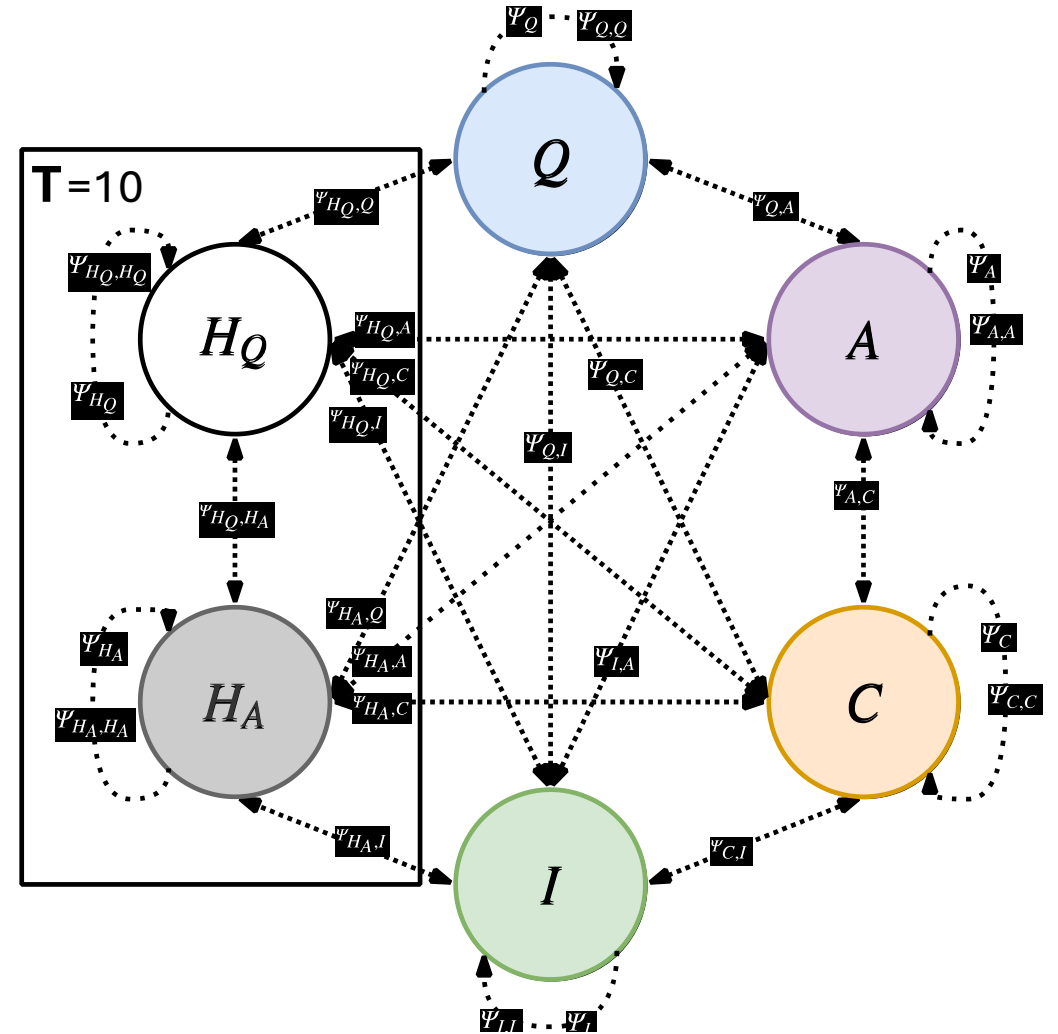
Caption

a group of kids around a table with a cake .



Factor Graph Attention

- **Node** - attention distribution over utility's elements
- **Factors** (i.e., Potentials) – model information and interactions between entities
- Aggregate information from utilities, which are neighbors in the graph



Any number of modalities: Messages and beliefs

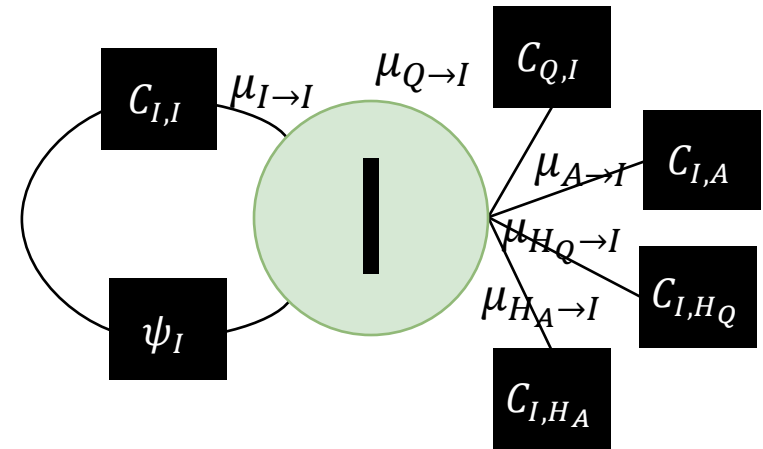
- We collect dependencies via

$$\mu_{Q \rightarrow I}(i) = \sum_{Q_j} W_{i,j} C_{Q,I}(i,j)$$

- We infer the amount of attention that should be given to each entity

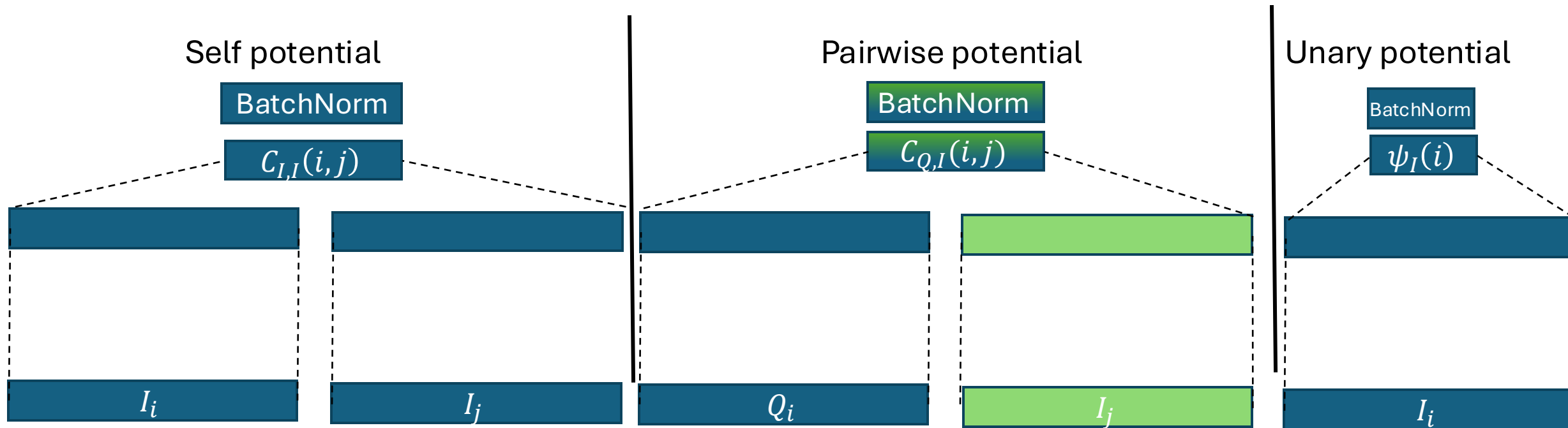
$$b_I(i) \propto \exp \left(w_I \psi_I(i) + \sum_{U \in \mathcal{U}} w_{I,U} \mu_{U \rightarrow I}(i) \right)$$

↑
set of modalities



Factors normalization

- We want to avoid situations where pairwise scores (e.g., image and question) bias another one (e.g., image and caption)
- we use a **batch normalization** for each factor output

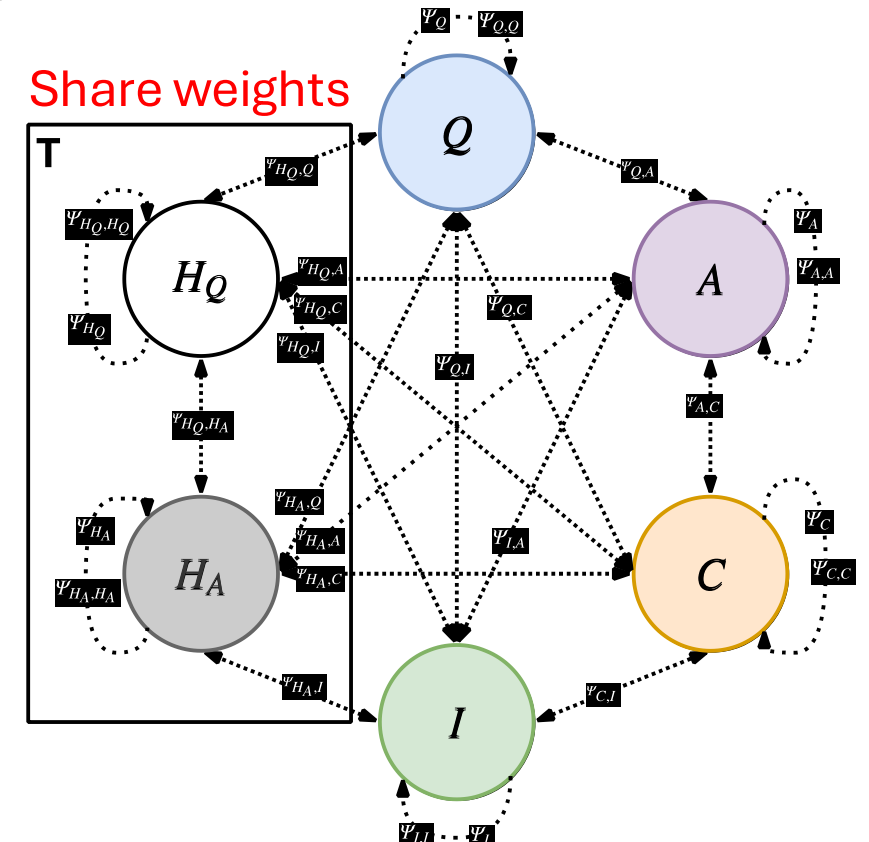


Weight sharing for similar interactions

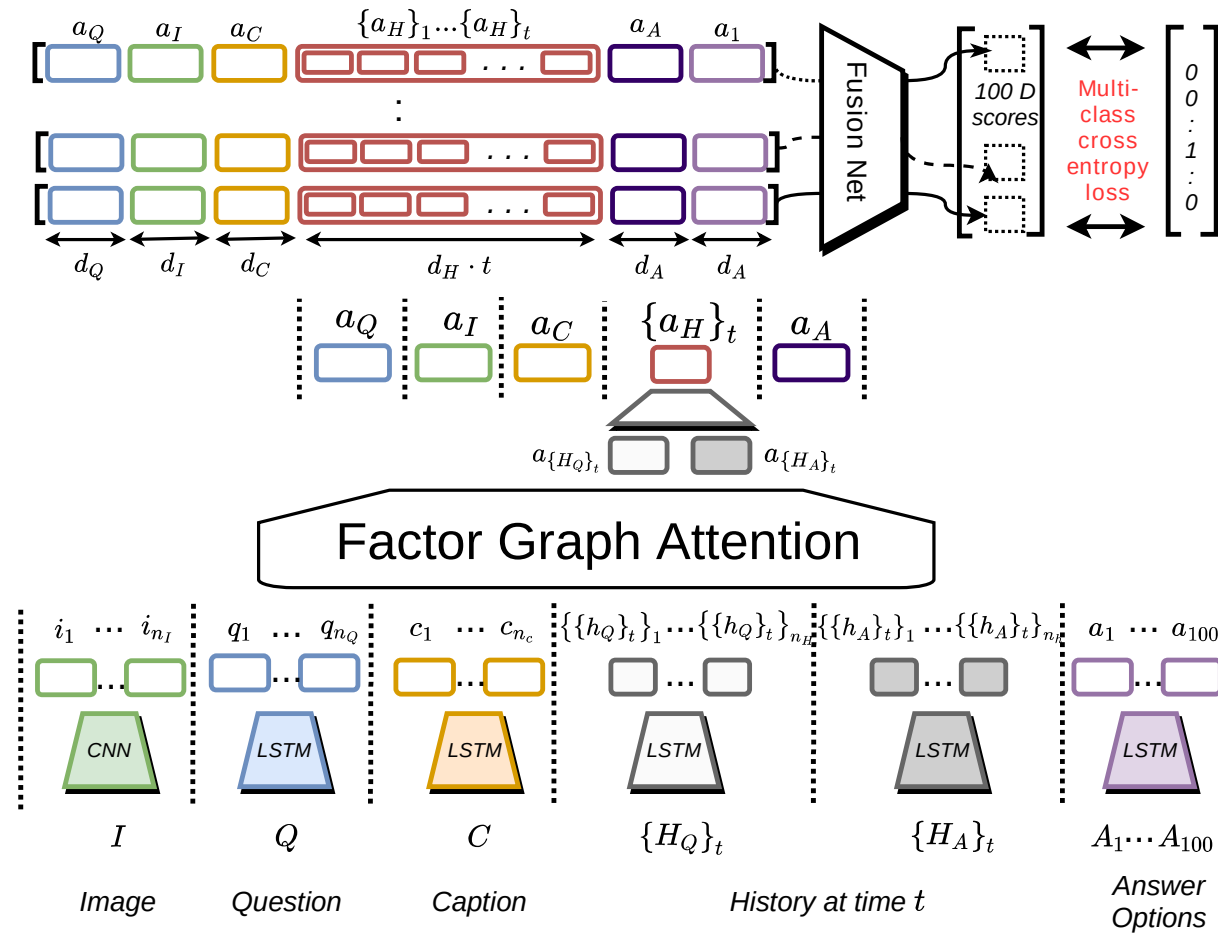
- In case of similar utilities, we can use the same interaction factor
 - e.g., History-Questions, and History-Answers

Reminder, pairwise interaction:

$$C_{A,B}(i,j) = \left(\frac{L_A A_i}{\|L_A A_i\|} \right)^\top \cdot \frac{R_B B_j}{\|R_B B_j\|}$$



State-of-the-art Visual Dialog architecture



Factor Graph Attention; Schwartz et al.; CVPR'19

Ensemble of MRR and NDCG models for Visual Dialog; Schwartz; (Winner of 2020 VisDial challenge)

Visual Dialog Results

Model	MRR	R@1	R@5	R@10	Mean
LF [10]	0.5807	43.82	74.68	84.07	5.78
HRE [10]	0.5846	44.67	74.50	84.22	5.72
HREA [10]	0.5868	44.82	74.81	84.36	5.66
MN [10]	0.5965	45.55	76.22	85.37	5.46
HieCoAtt-QI [33]	0.5788	43.51	74.49	83.96	5.84
AMEM [44]	0.6160	47.74	78.04	86.84	4.99
HCIAE-NP-ATT [32]	0.6222	48.48	78.75	87.59	4.81
SF-QIH-se-2 [21]	0.6242	48.55	78.96	87.75	4.70
CorefNMN [27]*	0.636	50.24	79.81	88.51	4.53
CoAtt-GAN-w/ $\mathbf{R}_{inte}$ -TF [53]	0.6398	50.29	80.71	88.81	4.47
CorefNMN (ResNet-152) [27]*	0.641	50.92	80.18	88.81	4.45
FGA (VGG)	0.6525	51.43	82.08	89.56	4.35
FGA (F-RCNNx101)	0.6712	54.02	83.21	90.47	4.08
Ensemble of 9 FGA (VGG)	0.6892	55.16	86.26	92.95	3.39

Video retrieval

Children and adults
are performing various
forms of martial arts



A reporter is talking
about a movie scene
from the wolverines



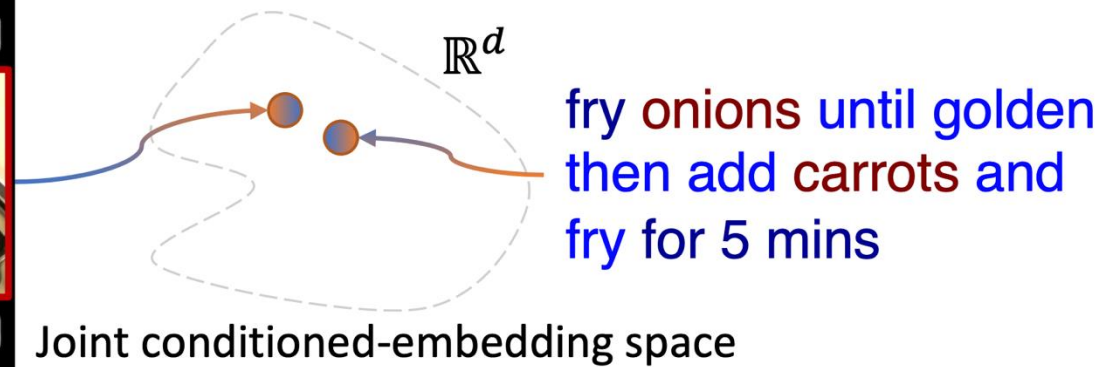
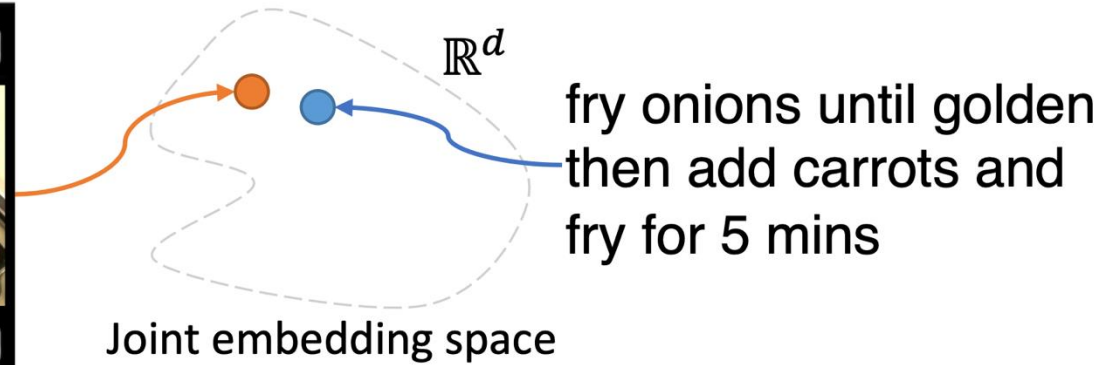
A man playing guitar
and a group of people
dancing with him



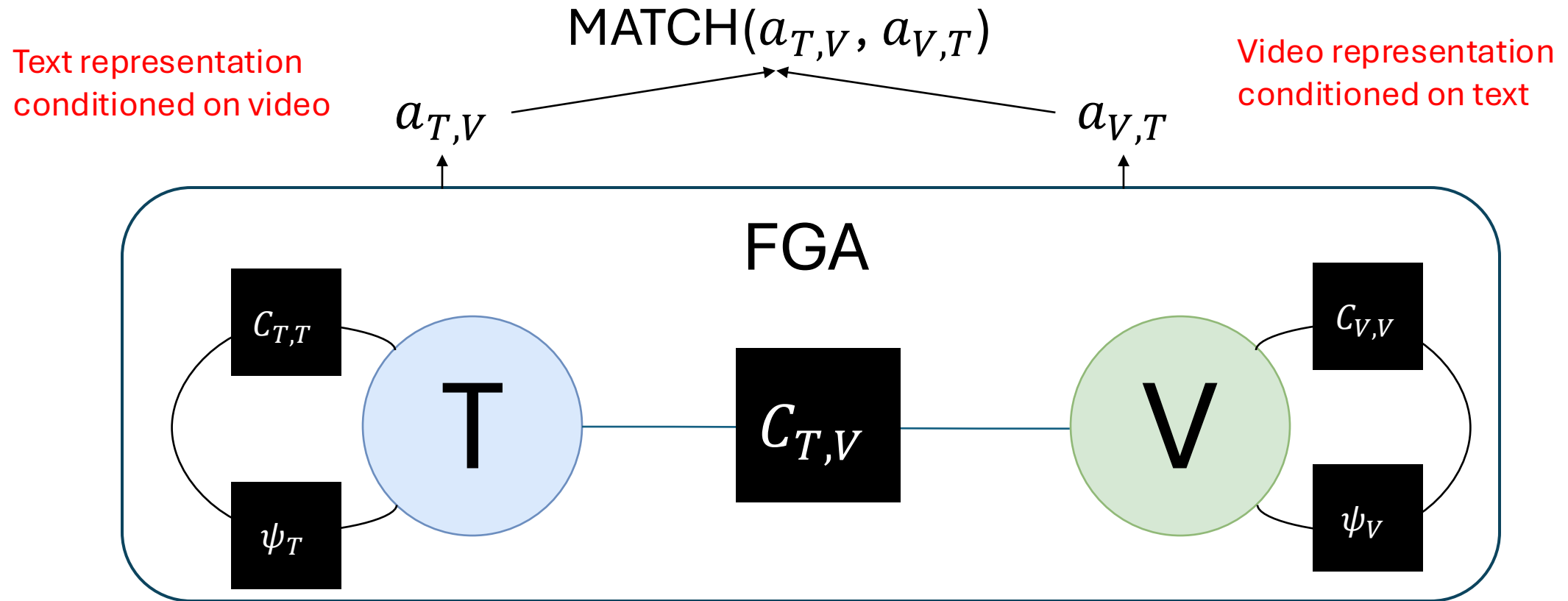
A person is melting
chocolate in a oven



Conditioned embedding space



Conditioned matching via Attention



A person is melting
Chocolate in a oven



The attention picks relevant-to-query frames and words

Low frame-attention belief



fry onions until golden then add carrots and fry for 5 mins

Low frame-attention belief



High frame-attention belief



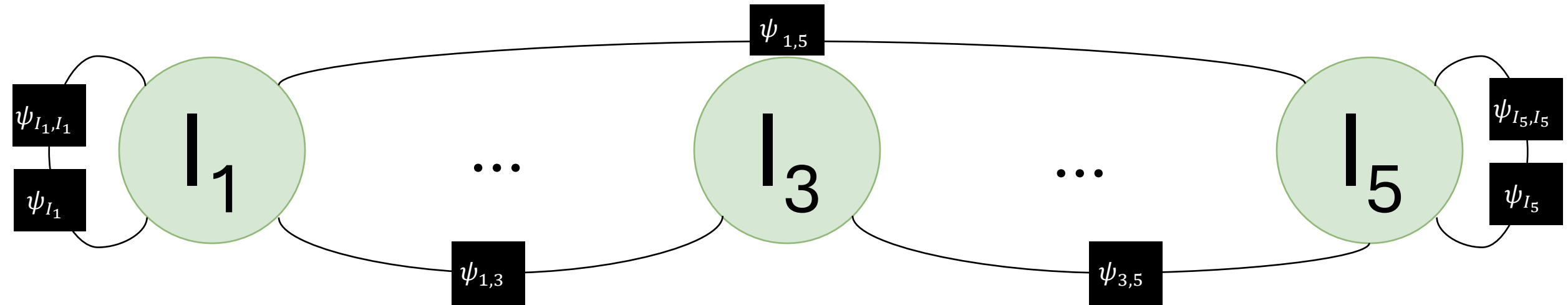
High frame-attention belief



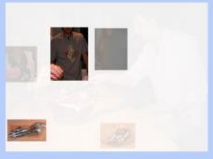
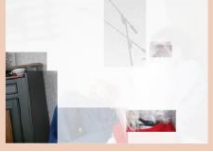
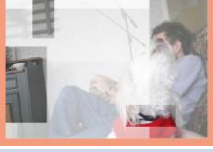
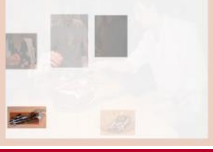
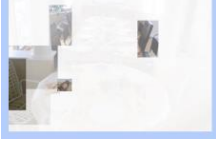
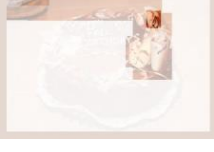
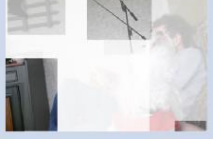
Two women are standing outside the bathtub spraying the dog with a hose.



Visual Storytelling



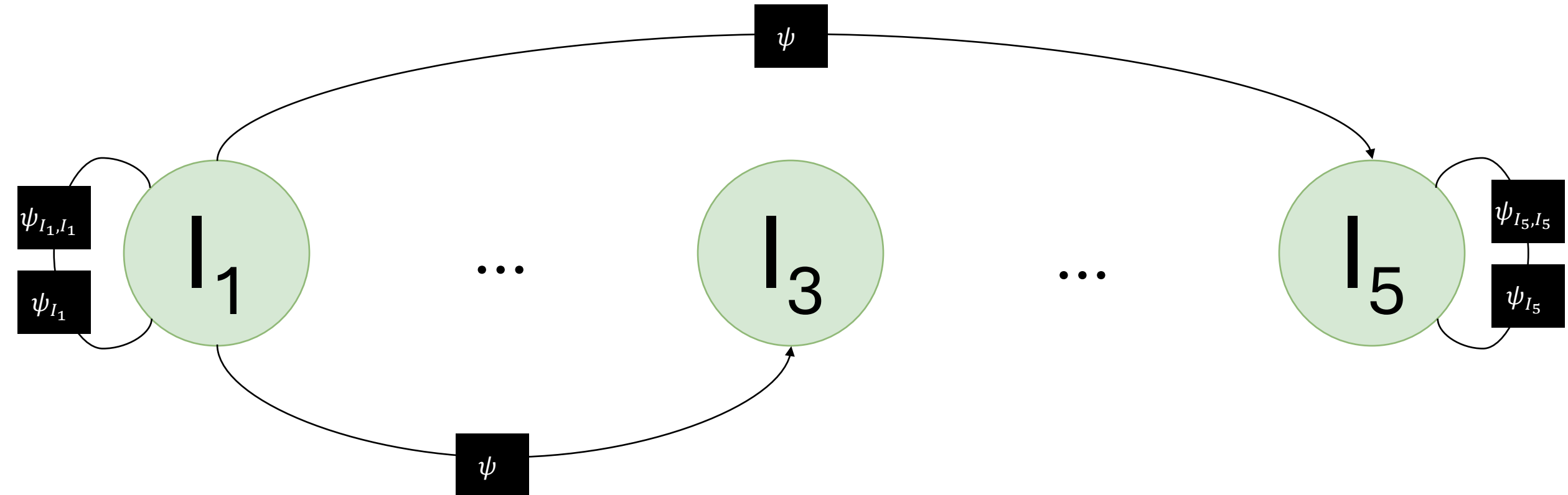
Attended representations to sentences

1.		→						→ The table was set for the party.
2.		→						→ The cake was delicious.
3.		→						→ We had a lot of food.
4.		→						→ The man is having a good time.
5.		→						→ We ended our night with a few drinks.

- The border highlights the important attended-image representations

Sharing-weights

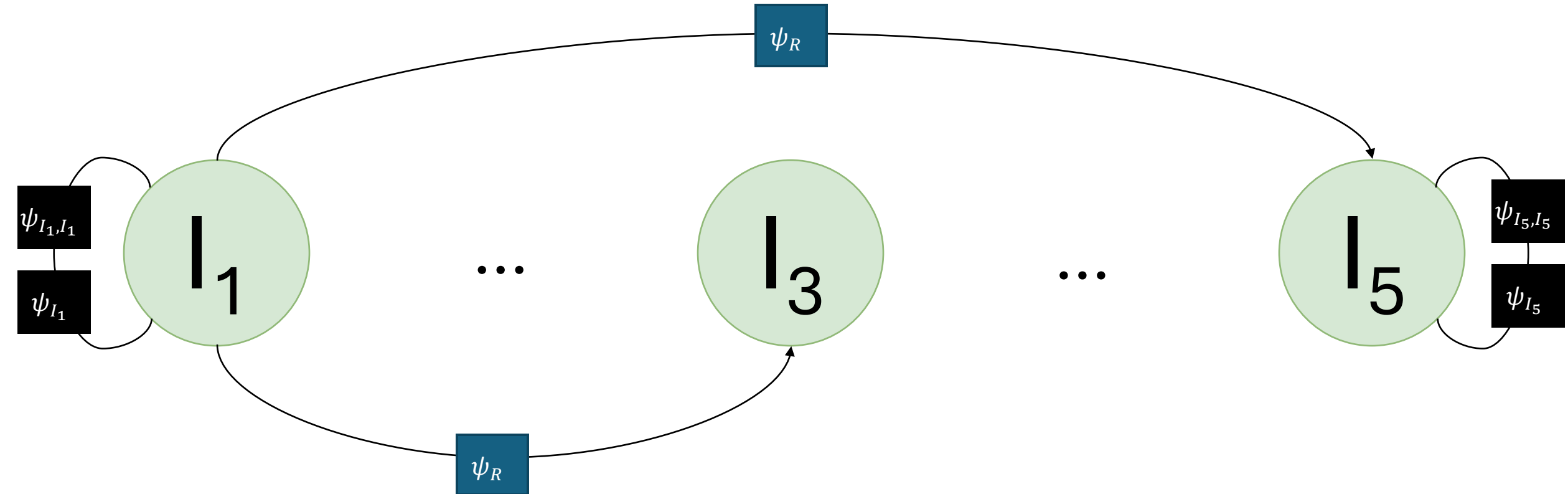
- Use the same factor for image interactions



- Does all interactions equal?

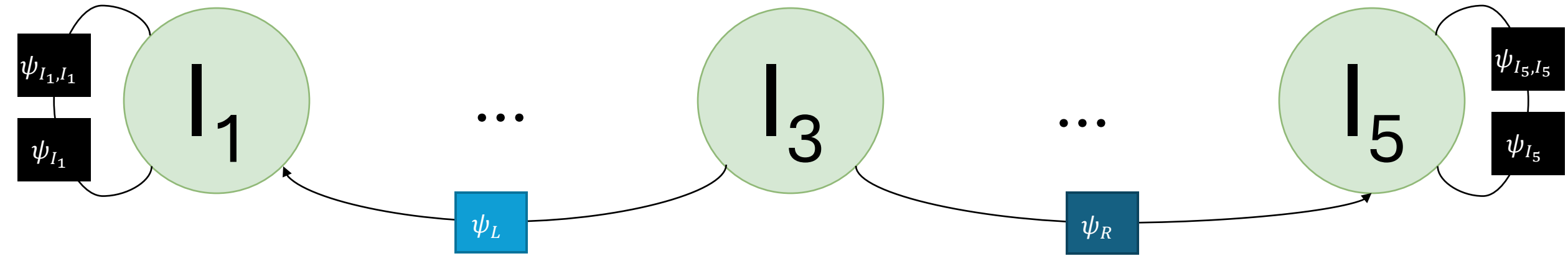
Order-aware factors

- 1st sentence



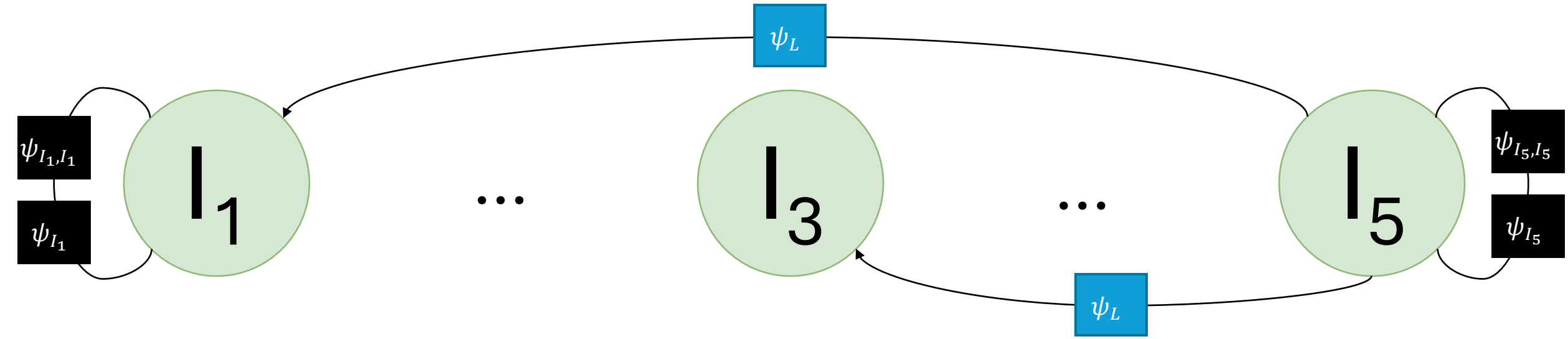
Order-aware factors

- 3rd sentence



Order-aware factors

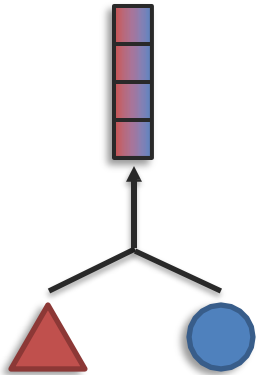
- 5th sentence



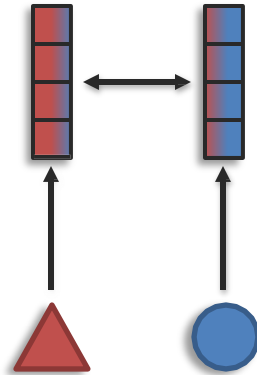
Summary: Representation

Sub-challenges:

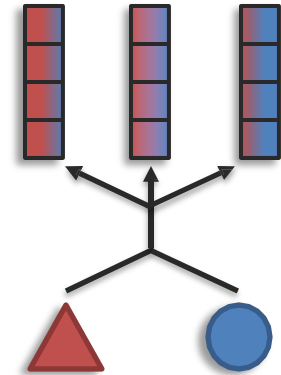
Fusion



Coordination



Factorization



Next week